The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Temporal-interference (TI) stimulation combines two high-frequency currents, yet their superposition contains no field component at the low difference frequency. The neural response requires field-to-membrane access, nonlinear detection, and amplification. We propose fast Single-neuron TI models already demonstrate axonal ion-channel rectification. We represent candidate fast access through axonal compartments, including nodes and the axon initial segmentand nodes, as candidate access sites. Evidence for periaxonal circuitry motivates treating axonal and somatic voltage responses separately; , by a short effective membrane time represents fast access in the population model. Its Periaxonal circuitry motivates separate axonal and somatic voltage responses. The field-to-voltage transfer and response at human field strengths require calibration. For weak perturbations of ongoing activity, smooth cellular nonlinearities with a nonzero second-order response generate a beat proportional to the product of the effective carrier amplitudes. A Jansen–Rit firing-rate sigmoid realizes this mixing through its curvature. An exact quadratic integrate-and-fire mean field connects the static curve to spiking dynamics: fast membrane relaxation leaves dynamic synapses, whereas outside that limit mixing depends on frequency. Membrane attenuation enters once, through access. Near a Hopf bifurcation, the network amplifies the generated beat as $1/\gamma$ within linear response, where γ is modal damping. Coupling sweeps separate this gain from leading-order beat generation. Rate modulation grows faster than the mean-rate change near resonance. Spiking simulations distinguish beat modulation below the oscillatory threshold from gamma induced when rectification crosses it; an equal-power mean-field carrier control separates induction from the beat response. The weak-input product law makes envelope-only dosimetry conditional on transduction. Within linear response, TI and direct low-frequency stimulation share the amplifier when coupling, readout, and state are matched, so resonance alone gives TI no efficiency advantage.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

Contents

1 Introduction **3**

2 Theory **4**
2.1 Population transfer and the slow-synapse limit 4
2.2 From applied field to a squared-envelope beat component 6
2.3 The exact mean field connects static and dynamic mixing 8
2.4 Transduction class and its spatial and spectral signatures 9
2.5 Amplification: the synaptic low-pass and the network resonance 10
2.6 Carrier access depends on the effective membrane response 11
2.7 Operating-point dependence 12

3 Methods **13**
3.1 Single-column Jansen–Rit: integration, lock-in, and controls 13
3.2 Laminar two-band model (LaNMM) 14
3.3 Exact next-generation mean field (NMM2 PING) 15
3.4 QIF spiking network (microscale of NMM2) 16
3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit) 17
3.6 Vocabulary: forced response, entrainment, and gating 17

4 Results **17**
4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input . . . 18
4.2 Near the Hopf, the demodulated beat drives a sharp resonance 18
4.3 Carrier independence and the kHz roll-off 19
4.4 The nonlinearity is the detector: the σ'' curvature law and its controls 20
4.5 Exact mean-field confirmation: the NMM2 PING gamma generator 21
4.6 What the mesoscale adds: coupling sets the gain at fixed detection 22
4.7 The resonance amplifies the envelope-locked modulation more than the mean rate 23
4.8 QIF responses distinguish beat modulation from field-induced gamma 24
4.9 Falsifiable predictions and tests 25

5 Discussion **26**

6 Conclusion **29**

A Notation and units **35**

B Simulated models: equations and parameters **37**

C The cubic Hopf normal form lacks a quadratic mixer **38**

D Efficiency of a memoryless transfer against a direct in-band drive **39**

E Extension: the laminar column at low carriers **41**

F Vocabulary and metrics **44**

G Supplementary figures **45**

1 Introduction

Temporal-interference (TI) stimulation was proposed to reach deep brain targets while reducing stimulation of surrounding tissue [1]. It applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs, typically with $f_1 \sim 1$–2 kHz and $\Delta f \sim 1$–100 Hz. Where the fields overlap, their superposition is an amplitude-modulated (AM) signal with an envelope at Δf. Electrode currents can steer the region of largest envelope modulation. Whether this also localizes the neural response depends on how neurons transduce the waveform.

The difference frequency is absent from the input itself:

$$s(t) = \varepsilon_1 \cos(\omega_1 t) + \varepsilon_2 \cos(\omega_2 t), \qquad \omega_{1,2} = 2\pi f_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta f. \tag{1}$$

Here $\varepsilon_{1,2}$ are the effective input amplitudes (related to the applied field in §2.2), Ω is the beat angular frequency, and $f_c = (f_1 + f_2)/2$ is the carrier frequency. The spectrum contains only the two applied lines (Fig. 3). A linear, time-invariant system can attenuate or phase-shift them but cannot create a new one. Passive membrane filtering therefore cannot demodulate TI [2]; a nonlinear element must mix the carriers into the low-frequency band, as in AM reception [3].

The mechanism therefore has distinct physical requirements (Fig. 1). *Access* is the conversion of the extracellular field into local membrane polarization. *Detection*, or demodulation, is the nonlinear generation of the difference-frequency component from those polarizations. *Amplification* is the response of the neuron and its recurrent circuit to the generated component. The same cellular structures can contribute to more than one stage, but strong coupling, effective mixing, and large resonant gain are different properties.

Access depends on the compartment that receives the field. A somatic charging time of 10–30 ms strongly attenuates kHz polarization [4, 5], whereas axonal geometry and myelination support different voltage responses. Cohen et al. [6] established a conducting periaxonal space beneath myelin that separates nodal and internodal voltage dynamics and supports saltatory conduction. This provides an anatomical basis for considering access through fast axonal compartments rather than assigning every compartment the somatic filter. Nodes and the axon initial segment (AIS) are candidate sites; the AIS also has a high density of voltage-gated sodium channels [7]. We represent this candidate route by a shorter effective membrane time. Its quantitative transfer from an applied TI field to local polarization remains to be calibrated; the measured and assumed time scales are distinguished in Table 2.

Detection requires that this polarization encounter a nonlinearity. Voltage-gated ion channels provide a cellular mixing mechanism in Hodgkin–Huxley and McIntyre–Richardson–Grill Mirzakhalili et al. [2] demonstrated subthreshold nodal mixing in a modified McIntyre–Richardson–Grill (MRG) myelinated-axon models [2]model with Hodgkin–Huxley ion-channel dynamics. Cortical-neuron models also show axonal excitation and conduction block under stronger fields [8]. Luff et al. [9] directly observed subthreshold difference-frequency oscillations in individual neurons, including carriers near 5 kHz. Mixing persisted during synaptic blockade but was suppressed by blocking tetrodotoxin-sensitive sodium channels. These measurements establish a cellular detector without identifying it with the proposed axonal access siteslocating it within a specific compartment. In a population description, smooth cellular contributions can be combined into an effective second-order response; a static rate curve realizes that response through its curvature.

Amplification determines whether the generated beat appreciably changes activity. Intrinsic conductances can tune a single neuron [10], as in the cell-type- and state-dependent TI responses reported by de Palma Aristides et al. [11]. Recurrent excitatory–inhibitory circuits provide a separate source of resonance [12, 13, 14]. In slices, pyramidal and parvalbumin (PV) neurons respond differently, and most pyramidal TI responses disappear under synaptic blockade [15]. Computational studies also find network-mediated responses [16, 17, 18]. Primate recordings at human-comparable fields report changes in spike timing without a corresponding mean-rate change,

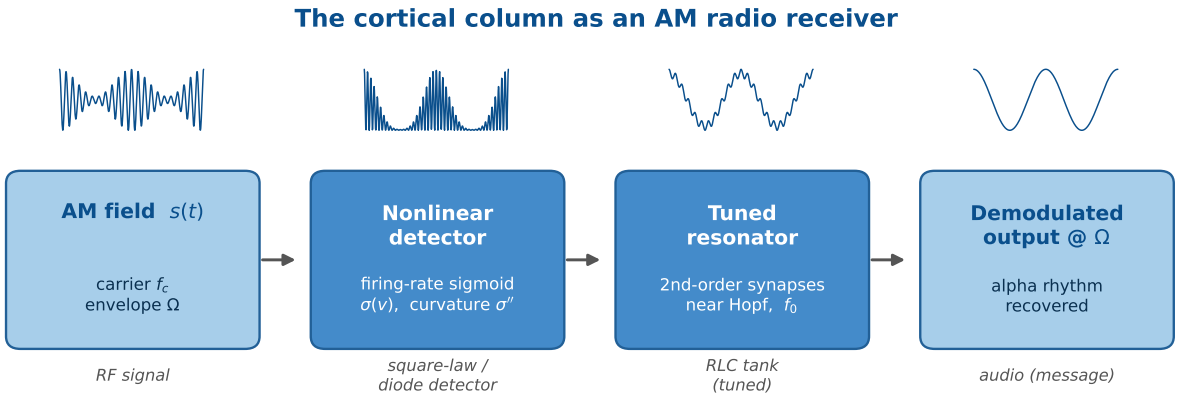


Figure 1: The cortical column as a tuned receiver. Access converts the applied field into the local polarization entering the detector. The nonlinear detector generates the beat at angular frequency Ω from a carrier near f_c; a recurrent network amplifies it near its natural frequency f_0. The diagram shows the static-sigmoid realization. In a conventional AM receiver, the tuned circuit selects the carrier before detection [3]; here the resonator selects the low-frequency output. The laminar extension also has a carrier-selective stage (§E).

with weaker effects than conventional low-frequency stimulation [19]. Cellular mixing and circuit amplification can therefore contribute to the same response.

We focus on weak perturbations of ongoing neural activity. Neurons may already be spiking; weakness refers to the stimulation-induced change around that operating state. A smooth response with a nonzero second-order term gives the leading beat, and linear amplification further requires stability under the rectified mean shift. Other preparations show carrier-driven excitation or block rather than envelope extraction [20, 21, 22]. Direct carrier-evoked spiking and conduction block lie outside the weak-input calculation; finite-amplitude network examples are identified separately.

The first calculation uses a Jansen–Rit firing-rate sigmoid: its curvature mixes the carriers, and a network with second-order synapses amplifies the beat near a Hopf bifurcation. An exact quadratic integrate-and-fire (QIF) mean field connects this static description to cellular dynamics. When membrane relaxation is fast relative to the input, a static population transfer curve emerges while the synapses remain dynamic; otherwise the mixing coefficient is frequency dependent. Simulations in the corresponding spiking network test the population description, and a laminar model extends the calculation to two resonant bands at low carriers.

Related neural-mass accounts use the same nonlinear mixing for endogenous cross-frequency coupling and nested amplitude modulation [23, 24]. Here the input is external, making the field-to-cell coupling an additional requirement. The proposed fast-compartment access route supplies the biological hypothesis; the population reduction connects its effective membrane response to detection Building on existing axonal TI mechanisms, we represent candidate fast access by a short effective membrane time and connect that response to population mixing and circuit amplification.

2 Theory

2.1 Population transfer and the slow-synapse limit

A neural mass model describes a population through averaged variables [25, 26]. In the Jansen–Rit (JR) family, a linear synaptic filter converts presynaptic firing rate $r(t)$ into postsynaptic potential $x(t)$:

$$\ddot{x} + 2a\dot{x} + a^2 x = Aa\, r(t),$$

with inverse synaptic time a and gain parameter A; inhibitory synapses use parameters b and B. Summed postsynaptic potentials give the mean input v, which a static transfer curve converts to firing rate [27],

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}}. \tag{2}$$

Here $2e_0$ is the maximum rate, v_0 the inflection voltage, and ρ the inverse voltage scale. Nonzero curvature makes this curve a mixer, while its slope sets the local gain of the recurrent circuit.

For a synapse from population Y to X, the resulting NMM1 equations are [25]

$$\hat{L}_{X\leftarrow Y}[u_{X\leftarrow Y}] = C_{X\leftarrow Y}\, r_Y, \qquad v_X = \Lambda[\hat{E}] + \sum_{Y:\, C_{X\leftarrow Y}\neq 0} u_{X\leftarrow Y}, \qquad r_X = \sigma(v_X). \tag{3}$$

The operator $\hat{L}$ is the inverse synaptic filter, $u_{X\leftarrow Y}$ its postsynaptic potential, $C_{X\leftarrow Y}$ the connectivity, and $\Lambda[\hat{E}]$ the direct polarization caused by the applied field. This field contribution enters alongside the synaptic potentials, without first passing through a synapse.

The corresponding population nonlinearity can be derived for all-to-all coupled QIF neurons with Lorentzian-distributed excitabilities. Their exact Montbrió–Pazó–Roxin (MPR) mean field evolves the firing rate r and mean voltage v [28]; adding dynamic synapses gives NMM2 [29, 30]. In membrane-time units,

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t), \tag{4}$$

where $\bar{\eta}$ is the mean excitability, $\Delta > 0$ its half-width, and $I(t)$ the normalized input, including recurrent drive when present. Rate and voltage are expressed in the corresponding rescaled units. For a constant total drive $Q = \bar{\eta} + I$, setting both derivatives to zero gives $v = -\Delta/(2\pi r)$ and

$$r = F(Q) = \frac{1}{\pi}\sqrt{\frac{Q + \sqrt{Q^2 + \Delta^2}}{2}}. \tag{5}$$

Equivalently, F averages the single-neuron steady rate $f(q) = \sqrt{q}/\pi$ for $q > 0$, and $f(q) = 0$ otherwise, over the excitability distribution. Here η is the neuron's excitability and $q = \eta + I$ its constant total drive. Threshold recruitment and the concave rate increase of active cells both enter its curvature. This nonlinear curve exists without recurrent coupling. It has an inflection at $Q = \Delta/\sqrt{3}$ and an unbounded concave rise, unlike the saturating JR sigmoid.

If the population relaxes to this equilibrium faster than synaptic inputs change, NMM2 reduces to a static rate map with dynamic synapses [29, 30]. Restoring the membrane time τ_m and physical firing rate, a second-order synapse with time constant τ_s obeys

$$r(t) \simeq \frac{F(Q(t))}{\tau_m}, \qquad \tau_s^2 \ddot{s}_{\mathrm{syn}} + 2\tau_s \dot{s}_{\mathrm{syn}} + s_{\mathrm{syn}} = r(t), \tag{6}$$

where s_{syn} is the filtered rate entering the recurrent drive in Q, with the original coupling normalization retained. This is the slow-synapse limit: shortening the effective membrane time makes the transfer appear static while the network retains its synaptic dynamics. The relevant separation is between the population relaxation time τ_{rel} and τ_s. Applying the same static map directly to a carrier of angular frequency ω_c requires the stronger condition $\omega_c \tau_{\mathrm{rel}} \ll 1$. A population can therefore be quasistatic for its synaptic inputs yet dynamic for a kHz carrier.

The transfer curve summarizes cellular nonlinearities within this regime. Its curvature combines their leading contributions to mixing; its slope converts a slow input into a rate change and enters recurrent gain. QIF supplies a derived realization through its spike-generating quadratic and excitability distribution. The generic weak-input mixing law does not require identifying each channel's contribution, although the coefficient and its frequency dependence are model-specific. Outside the static limit these contributions belong to a dynamic input–output functional. Section 2.3 calculates this response for QIF and recovers Eq. (5) in the adiabatic limit.

Cellular and circuit resonance are also distinct. Conductances such as I_h and I_M can tune single neurons [10]; the companion manuscript of de Palma Aristides et al. [11] reports cell-type- and state-dependent TI frequency selectivity in hippocampal neurons, sensitive to these conductances. Recurrent excitatory–inhibitory circuits can generate resonance without intrinsic cellular resonance [12, 13]. In the fast-membrane limit, any retained resonance belongs to the remaining synaptic circuit; modes requiring finite membrane dynamics need not persist.

The JR column couples pyramidal, excitatory-interneuron, and inhibitory-interneuron populations, with field input to the pyramidal sigmoid and local field potential (LFP) proxy $v = y_1 - y_2$ (App. B). The laminar extension (LaNMM) [23, 31] adds coupled alpha and gamma circuits: deep pyramidal P_1, spiny-stellate (SS), and somatostatin (SST) populations, and a superficial pyramidal P_2/PV circuit. These models retain dynamic synapses and static rate curves; NMM2 and its spiking realization retain membrane dynamics as well.

2.2 From applied field to a squared-envelope beat component

We use the quasistatic electromagnetic approximation for the applied field $E(t)$ (V/m), neglecting propagation and induction. This does not make membrane charging instantaneous. Let $s(t)$ be the polarization after the cellular access response; the static population model reads

$$r(t) = \sigma\big(v(t) + s(t)\big), \qquad E(t) = E_1 \cos \omega_1 t + E_2 \cos \omega_2 t. \tag{7}$$

The field acts directly on soma, dendrites, AIS, nodes, and terminals, each with its own coupling and frequency response. In a reduced description, their contributions to the effective input are represented by parallel entry points,

$$s(t) = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}(t), \qquad \boldsymbol{\lambda}_f^c = \boldsymbol{\lambda}_0^c \, |H_c(\omega_c)|, \tag{8}$$

where $\boldsymbol{\lambda}_0^c$ is the static polarization vector of compartment c (a length), H_c its access filter, and $\omega_c = 2\pi f_c$ [25, 32]. Equation (8) uses amplitude gains and assumes aligned compartment phases; a general sum requires the complex transfers, including their relative phases. The passive compartment equation $C\dot{\mathbf{V}} + A\mathbf{V} = \mathbf{I}^0 - \mathbf{p}E$ [29] gives $\delta\mathbf{V}(\omega) = -(A + i\omega C)^{-1}\mathbf{p}E(\omega)$, with capacitance matrix C, conductance matrix A, endogenous current $\mathbf{I}^0$, and field-to-current coupling $\mathbf{p}$. The static gain and its frequency dependence belong to this one response. A model driven by post-coupling polarization and a model driven by current through its own membrane equation must therefore count that filtering once.

For a dominant entry point, $\varepsilon_j = \lambda_f(\omega_j)E_j$ denotes the polarization reaching the nonlinearity. The direct-coupling calculation takes $\lambda_f \approx \lambda_0$ to isolate mixing and recurrent amplification; §2.6 restores frequency-dependent access. The proposed fast axonal compartments motivate Fast axonal access motivates this limit; their compartment anatomy enters the reduced model through the coupling and response time, which must be calibrated for the preparation.

Expanding σ about an operating point v^* (a Volterra/Taylor expansion, as in [24]) gives $\sigma(v^*+s) \approx \sigma(v^*) + \sigma'(v^*)\, s + \frac{1}{2}\sigma''(v^*)\, s^2 + \dots$ The truncation requires $|s| \ll 1/\rho \approx 1.8$ mV and a nonzero second derivative for quadratic mixing to be leading. Larger excursions require the full transfer curve. The linear term preserves the two input frequencies. The quadratic cross term generates their difference through the identity $2\cos\omega_1 t \cos\omega_2 t = \cos\Omega t + \cos(\omega_1 + \omega_2)t$. Averaging over the carrier (the fast $\cos^2 \omega_j t$ terms average to $\frac{1}{2}$, the sum-frequency term to zero),

$$\big\langle \sigma(v^* + s) \big\rangle_c = \sigma(v^*) + \tfrac{1}{4}\sigma''(v^*)\,(\varepsilon_1^2 + \varepsilon_2^2) + \tfrac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2 \cos\Omega t + O(\varepsilon^4), \tag{9}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon_1\varepsilon_2, \tag{10}$$

where $\simeq$ means to leading order in the polarizations, here and in what follows. The same expansion leaves a constant (DC) term, $\frac{1}{4}\sigma''(\varepsilon_1^2+\varepsilon_2^2)$, which any read-out of the mean rate must include.

For a dynamic cellular response with synaptic input held fixed, define the effective second-order coefficient $\chi^{(2)}$ by the complex beat amplitude $\widehat{r}_\Omega$,

$$\widehat{r}_\Omega \simeq \chi^{(2)}(\omega_2,-\omega_1)\,\varepsilon_2\varepsilon_1, \qquad r_\Omega(t)=\mathrm{Re}\left[\widehat{r}_\Omega e^{i\Omega t}\right].$$

Here $r_\Omega(t)$ is the generated rate component before recurrent feedback. With this cosine-amplitude convention, the memoryless limit is $\chi^{(2)}=\sigma''(v^*)/2$. Multiple cellular processes can contribute to the same coefficient. Small amplitude justifies the second-order expansion; it does not remove memory, so the coefficient may depend on both frequencies and the operating state. Symmetry or cancellation can make it vanish. QIF and the sigmoid realize this shared quadratic structure with different coefficients.

What the quadratic term recovers is not the envelope but its square. Writing $A(t)$ for the instantaneous envelope of the two-tone field, $A^2(t)=\varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos\Omega t$, the low-pass part of the squared signal is exactly

$$\mathrm{LPF}\left[s^2(t)\right]=\tfrac{1}{2}A^2(t)=\frac{\varepsilon_1^2+\varepsilon_2^2}{2}+\varepsilon_1\varepsilon_2\cos\Omega t. \tag{11}$$

This is square-law detection in the strict sense: the population transduces A^2, not A. The distinction matters because A^2 is exactly sinusoidal at Ω whereas A is not—at a matched target ($\varepsilon_1=\varepsilon_2=\varepsilon$), $A(t)=2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine.

The amplitude-modulated surrogate. The dense perturbative sweeps below use the three-line waveform

$$s(t)=\varepsilon\left[1+m\cos\Omega t\right]\cos\omega_c t, \qquad \omega_c=2\pi f_c, \tag{12}$$

with modulation depth m, because it parameterizes a sweep by one amplitude. It is not the physical signal: it carries a line at ω_c that the two-tone sum lacks. The two agree at Ω to leading order (the surrogate gives $A_\Omega=\frac{1}{2}\sigma''\varepsilon^2 m$) and are *not* equivalent beyond it. For matched genuine tones $\mathrm{LPF}[s^2]=\varepsilon^2(1+\cos\Omega t)$, whereas for the $m=1$ AM surrogate $\mathrm{LPF}[s^2]=\varepsilon^2(\frac{3}{4}+\cos\Omega t+\frac{1}{4}\cos 2\Omega t)$: the same Ω amplitude, but a different DC term and a spurious second harmonic at 25% of the fundamental. Since that second harmonic is the observable that distinguishes transduction classes (§2.4), every claim that turns on the waveform is run with genuine two tones: the demodulation demonstration (Fig. 3), the curvature and square-law controls (Fig. 6), the laminar maps (Figs. S1, S7) and the kHz demonstration. The AM surrogate is retained for the dense perturbative sweeps in (Ω,p) and (Ω,C), which read only the leading Ω component, and Fig. 6d verifies their open-loop agreement at one operating point. It does not establish quantitative equivalence of the recurrent sweeps near the Hopf, where the different DC terms can shift the operating state.

In the language of [24], the quadratic cross-term $s_1 s_2$ between a slow and a fast input (the second-order Volterra kernel) implements amplitude modulation/demodulation and the associated intermodulation lines. Demodulation is thus itself an intermodulation product: the line at Ω is the *difference frequency* of the two applied tones (ω_1 and ω_2), generated by the same quadratic term that, acting again on an intrinsic rhythm ω_0, produces the $\omega_0\pm\Omega$ sidebands seen at the spiking level (Fig. S12b). Equation (10) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega\propto\varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega\propto\sigma''(v^*)$, hence zero at the inflection $v^*=v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon=\lambda_f E_0$, so the field required for a given effect rises with f_c (§2.6)—consistent with the experimental consensus. Carrier independence is thus a statement about the recovered beat component at fixed polarization, not about the applied field.

2.3 The exact mean field connects static and dynamic mixing

The adiabatic condition in §2.1 determines whether the static curve describes carrier mixing. Consider an uncoupled MPR population (Eq. (4), membrane time set to one), with constant drive Q and a weak added current $a\cos\omega t$. To second order in a, the carrier-averaged rate shift is $a^2C(Q,\omega)$, where

$$C(Q,\omega)=\frac{r^*\,[\omega^2+3d^2-\nu^2]}{(d^2+\nu^2)\{[d^2+\nu^2-\omega^2]^2+4d^2\omega^2\}},\qquad r^*=F(Q),\quad d=\frac{\Delta}{\pi r^*},\quad \nu=2\pi r^*. \tag{13}$$

The fixed-point eigenvalues are $-d\pm i\nu$: d is the relaxation rate and ν the ringing angular frequency, both in membrane-time units. The limiting responses are

$$C(Q,\omega)\xrightarrow[\omega\to 0]{}\tfrac{1}{4}F''(Q),\qquad C(Q,\omega)\xrightarrow[\omega\gg d,\nu]{}\frac{F'(Q)}{2\omega^2}. \tag{14}$$

In the first limit the population tracks its static curve, whose curvature rectifies the input. In the second, the voltage excursion is approximately a/ω; its square supplies a slow drive that the population reads through F'. Above the inflection of F, these limits have opposite signs and the coefficient vanishes at $\omega_z=\sqrt{\nu^2-3d^2}$ (Fig. 7a). The 300 Hz and kHz carriers are on the fast side for the parameters of the pyramidal–interneuron network gamma (PING) circuit used below. That classification is parameter-dependent: a shorter effective membrane time can bring the same physical carrier into the adiabatic regime.

Membrane access is already included in this calculation. The linear voltage response is $H_v(\omega)=(i\omega+d)/[(i\omega+d)^2+\nu^2]$. Dividing by its squared magnitude gives the mixing coefficient at fixed voltage excursion,

$$K(\omega)=\frac{C(Q,\omega)}{|H_v(\omega)|^2}=\frac{F'(Q)}{2}\,\frac{\omega^2+3d^2-\nu^2}{\omega^2+d^2}. \tag{15}$$

This coefficient tends to $F'/2$ at high frequency. The $1/\omega^2$ suppression at fixed current comes from $|H_v|^2$, the membrane filtering already represented by Eq. (8); it is not an additional detector filter. A static curve placed after the same membrane has the same leading carrier roll-off, although its mixing coefficient and sign can differ. The steady-state curvature is $F''=4r^*(3d^2-\nu^2)/(d^2+\nu^2)^3$, so the inflection occurs at $\nu^2=3d^2$.

For the full eight-state PING (Eq. (27)), each high-frequency current tone of amplitude a_j produces a voltage phasor $a_j/(i\omega_j\tau_E)$. The v_E^2 term therefore supplies a difference-frequency source of magnitude $a_1a_2/(\tau_E^2\omega_1\omega_2)$ inside the equation for $\tau_E\dot v_E$. After division by τ_E, this source is propagated through the network susceptibility at Ω. At four stable operating points, the resulting response agrees with the full second-order calculation to within 1.1% at 300 Hz, 0.1% at 1 kHz, and 0.03% at 2 kHz (Fig. 7b).

The carrier and beat have different routes to the firing rate. At high frequency, the voltage equation integrates the input once, and the rate equation integrates the voltage again, giving a linear rate response proportional to $1/\omega^2$. The beat is generated in the voltage equation before this second integration and is read through the susceptibility at Ω. Thus its carrier dependence is $1/(\omega_1\omega_2)$, not the product of two linear rate responses. Among the quadratic terms in Eq. (4), v^2 contributes at order ω^{-2}, rv at ω^{-3}, and r^2 at ω^{-4} for nearby fast tones. Table 1 summarizes the comparison.

2.4 Transduction class and its spatial and spectral signatures

A neural response proportional to the squared envelope is not equivalent to one proportional to the envelope. A detector implements some transduction $F(A)$ of the local envelope $A(t)$, and the two laws predict different amplitude scaling, harmonic content and spatial weighting.

stage	static curve, same membrane	dynamic population
current → voltage	ω^{-1}	ω^{-1}
voltage → rate, carrier	1	ω^{-1}
current → rate, carrier	ω^{-1}	ω^{-2}
two tones → beat in the rate	$(\omega_1\omega_2)^{-1}$	$(\omega_1\omega_2)^{-1}\times$ const.

Table 1: Fast-carrier scaling with matched membrane access, at fixed beat frequency and a nonzero leading mixing coefficient. The dynamic population integrates the carrier once more between voltage and rate; the beat is formed before that step. Prefactors depend on the operating point and beat-frequency susceptibility.

For a narrowband field $s(t) = A(t)\cos\omega_c t$ passing through a memoryless nonlinearity g, the low-frequency (baseband) output Y_{LF} is the average of g over one carrier cycle:

$$Y_{\mathrm{LF}}(A) = (2\pi)^{-1}\int_0^{2\pi} g(v^* + A\cos\theta)\,\mathrm{d}\theta.$$

Odd powers of A average away, leaving only even ones:

$$Y_{\mathrm{LF}}(A) = g(v^*) + \tfrac{1}{4}g''(v^*)A^2 + \tfrac{1}{64}g^{(4)}(v^*)A^4 + \cdots \tag{16}$$

A smooth memoryless response with $g''(v^*) \neq 0$ therefore transduces A^2 at leading weak-input order, whatever its detailed shape. This includes the sigmoid mechanism of §2.2. Recovering A itself requires a qualitatively different detector: strong rectification. A full-wave rectifier gives $\mathrm{LPF}[\,|A\cos\omega_c t|\,] = (2/\pi)A$ and a half-wave rectifier $(1/\pi)A$, both linear in the envelope, as does a diode-and-capacitor peak detector. A hard threshold is not one of them: a step detector with offset δ transduces $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and nothing below it, which is neither linear nor quadratic in A (App. D). Asymmetric channel rectification, threshold crossing and release events are candidate biological realizations of rectifiers of one kind or another. Under a linear downstream response and fixed coupling, the amplitude sweep below distinguishes a quadratic effective drive from one proportional to the envelope; the derivation of §2.2 assumes a smooth transfer, and this section asks what would show that assumption to fail.

The two classes are distinguishable at a single site. At a matched target, $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos u| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1}\frac{(-1)^{n+1}}{4n^2-1}\cos 2nu$ gives a harmonic ladder in ratios $A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}A^2 = \varepsilon^2(1+\cos\Omega t)$ contains only DC and Ω. For a mixed detector $F(A) = c_1A + c_2A^2$ the pre-network ratio is

$$r_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_\Omega}\right| = \frac{1/5}{1 + \frac{3\pi}{4}\,c_2\varepsilon/c_1}, \tag{17}$$

so measuring r_{det} estimates $c_2\varepsilon/c_1$ directly.

Two effects make the harmonic ratio usable only with care, and the second decides how to run the test. First, a smooth detector does generate a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, so the square-law signature is a *vanishing* ratio in the weak-field limit, not an exact zero. Second, and more seriously, the recurrent network multiplies each harmonic by its own transfer gain, $r_{\mathrm{meas}} = r_{\mathrm{det}}\,|G(2\Omega)/G(\Omega)|$. A resonant column suppresses the second harmonic hard: driven at $\Omega = \omega_0$ with $\mathcal{Q} = 10$, a true 20% ratio reads as 0.7%, enough to make an envelope follower look square-law. The resonance is thus what would hide the discriminator, so the test must be run *off* resonance: at $\Delta f \simeq f_0/4$ both lines sit on the flat part of G, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ across $\mathcal{Q} = 3$–30, or with G measured independently by a weak sweep of direct low-frequency stimulation (transcranial alternating-current stimulation, tACS) and deconvolved.

An amplitude sweep tests the scaling rather than one ratio. An envelope-following detector gives $Y_\Omega \propto \varepsilon$ with the harmonic ratio pinned at 1/5; a weak square-law detector gives $Y_\Omega \propto \varepsilon^2$ with

a ratio growing as ε^2. The log–log slope of the envelope-locked response against field amplitude therefore separates the classes without requiring absolute calibration; the ε^2 control of §4.4 is that measurement, performed in the model.

Beyond harmonics and amplitude scaling, the transduction class fixes the spatial map. Define the excursion as $\mathcal{M}[y] \equiv \max_t y - \min_t y$ of a periodic quantity $y(t)$. The conventional TI envelope map is $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$, whereas the square-law detector responds to $\mathcal{M}[\frac{1}{2}A^2] = 2\varepsilon_1\varepsilon_2$, so the two are related pointwise by

$$\mathcal{M}\left[\tfrac{1}{2}A^2\right] = \max(\varepsilon_1, \varepsilon_2)\, \mathcal{M}[A]. \tag{18}$$

The square-law map is the conventional envelope-excursion map multiplied, at each point, by the dominant local polarization. It therefore does not preserve conventional TI focality. In typical montages, where the dominant carrier is substantially larger superficially than at depth, this reweighting degrades deep-to-superficial contrast: at $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially against $(0.5, 0.5)$ at a matched deep target, $\mathcal{M}[A]$ favors the superficial point by $2\times$ and $\mathcal{M}[\frac{1}{2}A^2]$ by $40\times$. The magnitude of the degradation is not fixed by this algebra and must be evaluated from the actual vector field distribution; a sufficiently weak second carrier can suppress the superficial product instead. The product law is not an account of TI focality.

The selectivity this mechanism does supply is dynamical rather than spatial. Writing $R_\Omega(\mathbf{x})$ for the demodulated response at the envelope frequency Ω produced at location $\mathbf{x}$, and $G_{\mathbf{x}}(\Omega)$ for the local network susceptibility there,

$$R_\Omega(\mathbf{x}) \;\propto\; \sigma''(v^*_{\mathbf{x}})\, G_{\mathbf{x}}(\Omega)\, \varepsilon_1(\mathbf{x})\, \varepsilon_2(\mathbf{x}), \tag{19}$$

a product of three factors. Field overlap sets where a difference-frequency drive is generated at all; the operating point and the network susceptibility set which of the driven populations amplifies it. Resonance does not override an unfavorable field product, but it can compensate for one. This is the radio analogy taken literally: received signal strength times receiver tuning.

Equation (19) is written for scalar projected fields. In general the coupling is vectorial and compartment-specific, $\varepsilon_i(\mathbf{x}) = \boldsymbol{\lambda}\cdot\mathbf{E}_i(\mathbf{x})$, so the mixing product $(\boldsymbol{\lambda}\cdot\mathbf{E}_1)(\boldsymbol{\lambda}\cdot\mathbf{E}_2)$ is signed: its magnitude follows the product of the projected amplitudes, while its phase inverts with either the sign of σ'' or the relative sign of the two projections. The derivations below take the collinear, scalar case.

2.5 Amplification: the synaptic low-pass and the network resonance

Synaptic filtering suppresses the carrier after the nonlinearity has generated the beat. Each JR excitatory synapse has transfer function $H_s(\omega) = Aa/(i\omega + a)^2$, a second-order low-pass with characteristic frequency $a/(2\pi) \approx 16$ Hz. It passes a 10 Hz component with relative amplitude about 0.7, while reducing a 100 Hz component to about 2.5% (Fig. 5). Provided the carrier does not itself excite a responsive mode, the loop therefore reads predominantly the generated low-frequency drive.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives a leading eigenvalue pair $\mu_\pm = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, with damping γ and modal frequency ω_0. Below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (10) at Ω gives a resonant response. A constant-numerator two-pole approximation gives

$$G(\Omega) \propto \left[(\omega_0^2 + \gamma^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0} \quad (\gamma \ll \omega_0), \tag{20}$$

so the response $\sim A_\Omega\, G(\Omega)$ peaks near ω_0, with an approximately Lorentzian power profile. Here ω_0 is the imaginary part of the eigenvalue; the canonical oscillator's undamped frequency is $\sqrt{\omega_0^2 + \gamma^2}$.

Its peak is at $\sqrt{\omega_0^2 - \gamma^2}$ when $\omega_0 > \gamma$. The full network transfer also contains frequency-dependent input and output projections, so this expression does not fix its exact peak. The leading $1/\gamma$ growth holds near a simple Hopf with nonzero projections, within the weak-drive regime. The rectified mean shift must also leave the operating state on the stable side of the bifurcation. Above the Hopf an autonomous limit cycle appears, and the demodulated drive can entrain it within an Arnold tongue; a large lock-in amplitude alone does not establish that phase capture.

Collecting the stages, the response at the beat is

$$R_\Omega \simeq G(\Omega;\gamma)\; D\; \lambda_f(\omega_1)E_1\, \lambda_f(\omega_2)E_2, \tag{21}$$

where λ_f maps field to local carrier polarization, D converts the product of those polarizations into a beat drive, and G maps that drive to the observed response. Their normalization depends on where the generated drive enters. For JR, $D = \frac{1}{2}\sigma''(v^*)$ and G is the response to a generated rate drive. For fast-carrier QIF in the normalized units of Eq. (4), $D = 1$ for the v^2 source and G includes its conversion through the voltage and rate dynamics. Thus the population readout is included once. In these limits D is independent of carrier frequency at fixed polarization; across the dynamic crossover it acquires frequency dependence. A carrier sweep and a beat sweep therefore constrain different stages rather than a single effective gain.

The resonance does not supply the quadratic mixer. The cubic Hopf normal form responds to an in-band drive but has no second-order difference-frequency source (App. C). A large resonant susceptibility alone therefore does not establish beat generation.

2.6 Carrier access depends on the effective membrane response

Carrier independence at fixed polarization does not imply independence at fixed applied field. To quantify access, consider a first-order membrane response $H_m(\omega) = 1/(1 + i\omega\tau_m)$, with effective time constant τ_m. For nearby carriers, the two polarizations acquire nearly the same attenuation. The quadratic beat then satisfies

$$\frac{A_\Omega(f_c)}{A_\Omega^{\text{direct}}} \simeq |H_m(\omega_c)|^2 = \frac{1}{1 + (\omega_c\tau_m)^2} \sim \frac{1}{(\omega_c\tau_m)^2} \quad (\omega_c\tau_m \gg 1), \tag{22}$$

where A_Ω^{direct} is the beat generated with the same static coupling and no access attenuation. This filter belongs either in the field-to-polarization map or in the membrane equation that computes that polarization.

Somatic passive time constants are commonly 10–30 ms; reported human pyramidal values include approximately 12 ms [4] and 16.5 ± 3.7 ms [5]. For the illustrative $\tau_m = 16$ ms used here, the power factor $|H_m|^2$ is about 10^{-4} at 1 kHz, 2.5×10^{-5} at 2 kHz, and 10^{-6} at 10 kHz. A static nonlinearity placed after this somatic response therefore receives strongly attenuated carriers.

Cohen et al.'s periaxonal circuit provides a reason to distinguish axonal from somatic access [6]. A conducting submyelin space, about 12 nm wide and partly sealed at the paranodes, separates the voltage across myelin from that across the axonal membrane. Their minimum fitted axonal decay was approximately 1 ms. This is a response to injected current in a distributed cable, rather than a measured transfer from a TI field. The AIS is another candidate site, with a high sodium-channel density [7]; this property supports nonlinear sensitivity but does not fix its charging time.

A shorter effective time changes carrier transmission quantitatively. The sensitivity case $\tau_m = 0.2$ ms gives $|H_m|^2 \approx 0.39$ at 1 kHz and 0.14 at 2 kHz (Fig. S5). This assumed time represents the fast-compartment hypothesis in Eq. (8), or motivates a shorter membrane time in a QIF reduction; it has not been measured under TI at the AIS or a node. In the latter description, the approach to a static transfer is governed by Eq. (6); the membrane filter must not be applied again outside the QIF equations. A short access time improves carrier transmission without, by itself, specifying the mixing coefficient or the remaining circuit resonance.

Response	Time scale	Evidence and role
Somatic access	$\tau_m = 16$ ms; $f_{\text{char}} \approx 10$ Hz	Illustrative filter, informed by somatic measurements near 12 and 16.5 ms [4, 5].
Myelinated axon	minimum fitted decay ≈ 1 ms	Injected-current responses in Cohen et al.'s periaxonal circuit [6]; evidence for distinct axonal voltage dynamics.
Fast axonal access	assumed $\tau_m = 0.2$ ms; $f_{\text{char}} \approx 800$ Hz	Reduced-model sensitivity case for a fast compartment, such as an AIS or node; TI coupling and charging time require calibration.
Field-induced charging	≈ 0.7 μs	Measured with strong microsecond field pulses in isolated neurons [22]; a different stimulation regime.
Cellular detector	channel- and state-dependent	Luff et al.'s TTX-sensitive mixing persists near 5 kHz [9]; this measures detection, not a passive time constant.
JR excitatory synapse	$\tau_s = 10$ ms; $f_{\text{char}} \approx 16$ Hz	Model filter acting on the generated rate, downstream of detection.

Table 2: Access, detection, and downstream filtering involve different response times. Measured decays, pulse-charging times, and assumed model parameters are identified separately. The characteristic frequency $f_{\text{char}} = 1/(2\pi\tau)$ is given only for the specified filters; an axonal decay is not automatically a TI cutoff. AIS, axon initial segment; TTX, tetrodotoxin.

Geometry can also alter the relevant response mode. For a passive cable, the magnitude of the frequency-dependent space constant is

$$\ell(\omega) = \ell_0(1 + \omega^2\tau_m^2)^{-1/4}, \tag{23}$$

where ℓ_0 is its static value. At 1 kHz and $\tau_m = 16$ ms, this length is about one tenth of its static value, so compartments that are strongly coupled at low frequency can respond differently to the carrier. A somatic charging time alone therefore does not determine every extracellular-field response mode. These distinct modes motivate the compartment-specific access hypothesis; their implications for TI are considered in §5.

Channel kinetics contribute to the effective dynamic mixing response. A gate filter added outside the detector must not duplicate dynamics the detector already contains, and where the detector lacks them, as QIF lacks sodium-channel kinetics, the filter needs its own model or measurement; neither a static sigmoid nor QIF specifies a universal 1 kHz gate cutoff. Equation (13) gives the QIF realization of this response. Carrier sweeps can constrain the combined access and mixing response, while beat-frequency sweeps probe the downstream amplifier.

2.7 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 6). The Population Excitation–Inhibition Index (PEIX) of [23] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\text{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \text{sech}^2\left[\tfrac{\rho}{2}\,(v^* - v_0)\right] \in (0, 1], \tag{24}$$

a dimensionless read-out of where v^* sits, the side given by $\text{sign}(v^* - v_0)$. PEIX thus tracks σ' (maximal at the inflection v_0), whereas the demodulation gain tracks σ'' (zero at v_0, peaked on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks where the demodulation gain vanishes.

This complementarity is a property of the memoryless detector. In the exact mean field at a fast carrier the rectified DC term follows the slope F' and is largest at the inflection (§2.3), so the relation between PEIX and TI efficacy is model-dependent; PEIX remains a measurable read-out of the operating point. In the Jansen–Rit detector, anything shifting v^* (neuromodulation, attention, arousal, or pathology such as psychedelics or PV-interneuron dysfunction in Alzheimer's disease) scales, and can flip the sign of, the TI response; in the dynamic detector the same shifts act through F' and the susceptibility, without a sign rule. The state dependence is the falsifiable handle; the sign reversal is specific to the memoryless model.

3 Methods

3.1 Single-column Jansen–Rit: integration, lock-in, and controls

All symbols and units are collected in Table 4 (App. A). We integrate the single-column JR model of §2.1 (Eqs. (28), parameters in Table 5) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (28), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at Ω in v is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta (RK4) scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S9 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and 1.5×10^{-5} s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.6 are analytic evaluations of Eq. (22), not simulations.

The resonance sweep in Fig. 4 uses 10 s of settling and a 20 s measurement window. Long windows reduce transients from weakly damped modes and average beating when autonomous activity persists. Other runs use the windows specified in their scripts. The two-tone 2 kHz example (Fig. S6) uses $p = 400$ s^{-1}, $C = 135$, and 0.5 mV per tone with $\tau = 0.2$ ms. From a zero initial state, it settles for 4 s and records 4 s at $\Delta t = 1.5 \times 10^{-5}$ s, sampling before each RK4 step. Its amplitude spectra use plain-mean subtraction and a Hann window. Parameter sweeps are vectorized: the state is an $(N, 6)$ array advanced simultaneously for all N parameter combinations, so a full resonance map runs in seconds.

The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically and diagonalizing it: the Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\mu_+$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \mathrm{Im}\,\mu_+/2\pi \approx 11.1$ Hz—consistent with the continuation and with the JR bifurcation analysis of [33]. For $p > 315$ the fixed point is a stable focus (resonator); between the saddle-node on an invariant circle (SNIC, $p \approx 114$) and the Hopf the column free-runs on the alpha limit cycle.

The response at the envelope frequency is read out with a software lock-in amplifier referenced to Ω—a standard way to isolate the component of a signal at one chosen frequency: multiply by sine and cosine at Ω and average, so averaging suppresses other frequencies and estimates the component at Ω. A finite window does not eliminate neighboring frequencies exactly. We multiply $v(t)$ by quadrature references and integrate over a Hann-windowed measurement window to obtain the in-phase (I) and quadrature (Q) components,

$$I = \frac{2}{\sum_k w_k} \sum_k w_k\,(v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k\,(v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2}, \quad (25)$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to -A\sin\phi$, so $A_\Omega \to A$; the

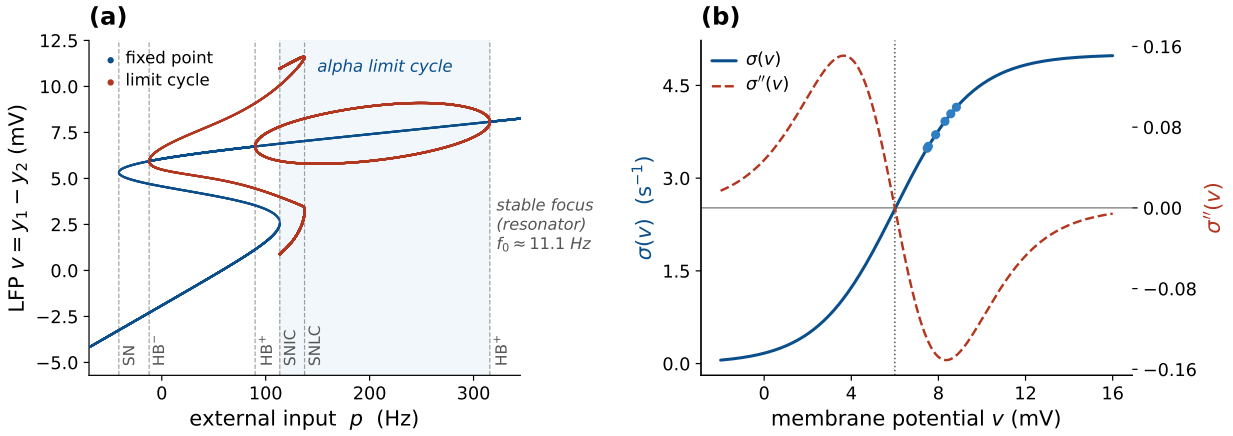


Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature. *(a)* JR bifurcation diagram from numerical continuation—the LFP $v = y_1 - y_2$ along the fixed-point branch (blue) and the min/max of the limit-cycle branches (red) vs. external input p, with *solid* stable and *faded* unstable. The alpha limit cycle (shaded) is bounded by a saddle-node-on-invariant-circle (SNIC, $p \approx 114$) and the supercritical Hopf ($p \approx 315$, $f_0 \approx 11.1$ Hz); beyond the Hopf the fixed point is a stable focus, the near-Hopf resonator used here. Further codim-1 points are marked (saddle-node SN, Hopf HB$^{\pm}$, saddle-node of limit cycles SNLC). *(b)* The sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection v_0. Dots mark the operating points reached by the closed-loop model for the p values used here.

factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. Out-of-band components can leak into the estimate, particularly the DC level of v ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). The finite-window projection of a constant onto the quadrature references need not vanish, so we subtract $\bar{v}$ explicitly; this makes the metric accurate down to the 10^{-8} level needed for the kHz analysis (§2.6). For the open-loop rate readout, the signed in-phase component I recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 6).

We compare the *open-loop detector*, the firing-rate sigmoid at a fixed operating point v^* without recurrence, with the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the two-tone field through σ at v^* and applies Eq. (25), isolating the pure demodulation law (Figs. 5, 6, S5). The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in Eq. (28) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against finite differences. Code and reproduction limits are listed under *Code and data availability.*

3.2 Laminar two-band model (LaNMM)

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [23, 31], with the field entering as in the single JR column, as a membrane polarization added inside the sigmoid argument of the target population,

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\}, \tag{26}$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude ε in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S2). A perturbation of an external input *rate* would require clipping at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at Δf into the drive before the population nonlinearity received it. A signed membrane polarization

carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, ε is directly comparable between the two models.

Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid simultaneously, as in the JR sweeps. The maps report the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials as the mean squared magnitude of the Hann-windowed fast Fourier transform (FFT) over the band, in arbitrary units, over a post-transient window. This measure includes autonomous and driven activity and does not isolate a phase-coherent response to stimulation.

Two sweeps probe the column: an amplitude–beat sweep over $(\varepsilon, \Delta f)$ at fixed carrier, and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed ε, built on the LaNMM module of [23]. The maps (Figs. S1, S7) show the *signed* stimulation-induced change, alpha power minus the unstimulated ($\varepsilon = 0$) baseline, on a zero-centered scale; over most of the P_1-driven grids the change is negative. For the alpha resonance we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (25)) of v_{P_1} at Δf.

The additional carrier comparison in Appendix E uses the lock-in of v_{P_1}, averaged at each carrier over 17 beats: 7–9 and 12.25–14 Hz in 0.25 Hz steps. These lie more than 1.5 Hz from the field-free alpha peak near 10.6 Hz. We denote this arithmetic mean by $S(f_c)$. The carrier grid is 25–80 Hz in 1 Hz steps, with equal tone amplitudes $\varepsilon = 1.5$ mV for P_2 drive and 0.5 mV for P_1 drive. After 2 s of settling, we compare 3 and 10 s measurement windows. Field-free controls give mean background amplitudes of 0.0048 and 0.0012 mV over the same beats. This background includes a persistent component near 8 Hz, so it is not solely finite-window leakage. The statistic compares carrier profiles within each drive condition; the unequal amplitudes do not support a direct sensitivity comparison between targets.

3.3 Exact next-generation mean field (NMM2 PING)

We test the mechanism in the exact next-generation mean field (NMM2, §2.1), following the Rosetta Stone formulation of [25] (which places the Montbrió–Pazó–Roxin reduction [28] on a common footing with the JR and LaNMM models used here), where the firing-rate nonlinearity is the derived MPR dynamics of Eq. (4) rather than a postulated sigmoid. We use a two-population excitatory–inhibitory (PING) motif [29, 30, 25]: each population $\alpha \in \{E, I\}$ is an MPR mean field (firing rate r_α, mean potential v_α) with membrane time τ_α, coupled through second-order synapses,

$$\begin{aligned} \tau_E \dot{r}_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, & \tau_E \dot{v}_E &= v_E^2 + \bar{\eta} - (\pi\tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + I_E(t), \\ \tau_I \dot{r}_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, & \tau_I \dot{v}_I &= v_I^2 + \bar{\eta} - (\pi\tau_I r_I)^2 + \tau_I C\,(s_E - 2 s_I), \end{aligned} \tag{27}$$

with second-order synaptic activations $s_E = \hat{K}_a[r_E]$ (excitatory, time constant τ_a) and $s_I = \hat{K}_g[r_I]$ (inhibitory, τ_g), each obeying the same critically-damped operator as the JR synapse, $\tau^2 \ddot{s} + 2\tau \dot{s} + s = r$; here C is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows: the E membrane sees $C\,(s_E - s_I)$ and the I membrane $C\,(s_E - 2 s_I)$, the only asymmetry being the doubled $I \to I$ weight $A_{ii} = 2$.

The common background drive $\bar{\eta}$ is the bifurcation parameter: raising it past a supercritical Hopf at $\bar{\eta}_H = 1.110$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory voltage equation as a current-like forcing, $I_E(t) = P\,E(t)$ in the convention of the exact mean field [29], with the dipole coupling P instantaneous, as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) of amplitude A_f in the dimensionless MPR units of v_E. The E membrane $\tau_E \dot{v}_E$ integrates this forcing into the carrier-frequency excursion $A_f/(\omega\tau_E)$ that the exact quadratic v_E^2 term rectifies, with no postulated sigmoid; the PING as run therefore computes membrane access with $\tau_m = \tau_E$.

We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (25)) of r_E at Δf over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [25]): $\Delta = 1$, global coupling $C = 15$,

$\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz), located by bisection on the leading eigenvalue of the mean-field Jacobian. The simulations retain these finite membrane times; Eq. (6) characterizes the fast-membrane limit.

Two parameters reach this gamma Hopf, and we use both as criticality knobs—just as the external drive p and the coupling C both reach the JR alpha Hopf. The background drive $\bar{\eta}$ (above) is swept for the resonance map (Fig. S8); the global coupling $C\,(\equiv J)$ is swept for the J-curve (Fig. S10): we vary J at fixed $\bar{\eta}$ toward the gamma Hopf J^* (located by bisection on the leading Jacobian eigenvalue) and read the lock-in peak of r_E over Δf. Unlike the JR J-curve, this sweep does not pin the operating point. In the fast-carrier approximation the leading voltage source is fixed by the carrier excursion (§2.3). The four-state check bounds the error of that approximation at the tested points; it does not establish source invariance over the full J sweep. The plotted normalization therefore uses the leading fast-carrier source, while the finite-carrier response can retain operating-state dependence.

We check the stability of the driven J-curve endpoint by integrating the full variational matrix along its trajectory. At carrier 300 Hz and beat 50.5 Hz, the AM input repeats every 2 s. After 8 s of settling, we compute the perturbation multipliers over that period and repeat over the next period. Moduli below one indicate decay of perturbations to the periodic response.

3.4 QIF spiking network (microscale of NMM2)

NMM2 is the exact infinite-population mean field of all-to-all QIF neurons [28, 25, 30]. We integrate that network directly to compare beat-locked and ongoing activity at the spike level (§4.8): $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (27)), with finite spike threshold $V_{\text{peak}} = 100$ and reset $-V_{\text{peak}}$ in model voltage units, excitabilities $\eta_j \sim \text{Lorentzian}(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout.

The common background drive $\bar{\eta}$ places the field-free mean field below or above a gamma Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz). We use $\bar{\eta} = 0.85$ with a 55 Hz beat and $\bar{\eta} = 1.5$ with a 42 Hz beat. Both runs have carrier $f_c = 300$ Hz and amplitude $A_f = 35$ per tone. These are finite-amplitude illustrations: the fast-carrier estimate gives a voltage excursion $A_f/(\tau_E\omega_c) \approx 1.2$ per tone in the model's voltage units. The field can therefore alter the dynamical regime, even when the unstimulated state is below Hopf. We label the conditions by their field-free state and separate the beat-locked component from other gamma activity. The spectra and rate folds do not by themselves establish phase capture (§F).

We also run both conditions at $A_f = 10$ per tone with the same measurement settings. For $\bar{\eta} = 0.85$, separate checks double the population size, double the measurement window, or halve the integration step; further runs raise V_{peak} to 200 and 400 at both time steps. An equal-RMS mean-field control replaces the two tones by one 300 Hz carrier with peak amplitude $\sqrt{2}A_f$, using the same duration: both inputs have mean-square amplitude A_f^2 over these windows. The carrier comparison starts from the field-free equilibrium and uses 800 ms settling and 800 ms measurement, with a 3200 ms window to check spectral leakage at the larger amplitude.

For two nearby carriers, the fast-carrier approximation gives a constant shift of the excitatory drive, $\delta\eta_E \simeq A_f^2/(\omega_c\tau_E)^2$, from the quadratic voltage term. We refine this estimate using the full PING linear carrier responses and their second-order DC source, then solve for the carrier-averaged equilibrium and its stability, including the source's dependence on that equilibrium. This approximation isolates the mean shift; direct two-tone integrations retain the beat forcing and provide a separate check.

The network is integrated with Euler steps of 0.0025 ms, settled for 800 ms (over four field-free

relaxation times at $\bar{\eta} = 0.85$, where the mean-field relaxation time is 176 ms) and measured over an integer number of beat cycles, 44 at 55 Hz and 34 at 42 Hz (800 and 810 ms), removing the phase-coverage bias from incomplete beat cycles. Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega/\langle r_E \rangle$ (a stable measure; the off→on lock-in ratio would divide by the field-free floor), and per neuron by the pairwise phase consistency of each neuron's spike phases at Δf [34], which is unbiased in the spike count, over neurons with at least ten spikes in the window.

In the $A_f = 35$ convergence checks, halving the time step changes the lock-in and the pooled vector strength by under 1%; doubling the spike-detection threshold changes the lock-in by about 1% and the mean rate by 2%, but the pooled vector strength at $\bar{\eta} = 0.85$ by about 45%, so phase concentration, unlike rate modulation, is sensitive to how spike times are defined.

3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The coupling sweeps of Figs. 9 and S11, and the transcranial alternating-current stimulation (tACS) control of Fig. S3, use the single JR column with the field-free operating point v^* pinned: for each global connectivity C we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds v^* (hence $\sigma''(v^*)$) fixed, vary $C\,(= J)$ toward the alpha Hopf, and read γ, ω_0 from the 6×6 Jacobian. The field is the AM surrogate of Eq. (12) for TI (Fig. 9; AC lock-in vs DC mean rate for Fig. S11) or a direct in-band sinusoid $s(t) = \varepsilon \cos \Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S3), each read out by the lock-in of v at Δf. The Figure 9–Figure S3 response-ratio comparison uses $v^* = 8.30$ mV, TI surrogate amplitude 0.3 mV with $m = 1$ and $f_c = 100$ Hz, and tACS amplitude 0.1 mV, driven at each field-free modal frequency. The TI amplitudes are interpolated logarithmically in C onto the tACS grid. Pinning fixes the background curvature; it does not cancel the stimulation-induced mean shift. Additional controls at $C = 110$, 133.9091, and 136.3 use a tACS amplitude of 0.0143039 mV, matched to the leading TI beat source. They record mean voltage, mean rate, and lock-in with 12 s settling and a 6 s window, repeated with 60 s settling and a 20 s window.

3.6 Vocabulary: forced response, entrainment, and gating

Forced response denotes the component driven at the beat Δf, quantified by the lock-in amplitude (Eq. (25)). A purely forced regime requires a stable operating state under stimulation; being below Hopf with the field off is insufficient. *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator [35]; in the maps we screen for it with the dominance criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency. This identifies frequency dominance, not phase capture. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [36, 37]. Capture and gating are dynamically different, and the same word would otherwise cover both. The QIF rate fold measures beat-locked firing, which does not by itself establish gating of gamma amplitude. Appendix F gives the metrics and the frequency-dominance protocol.

4 Results

The Jansen–Rit column and the exact next-generation mean field (NMM2) both generate a difference-frequency component and amplify it near a Hopf bifurcation. The former uses a memoryless sigmoid; the latter derives the nonlinearity from QIF dynamics. Their detector coefficients differ, but both separate beat generation from recurrent gain. Coupling controls that gain at fixed generated drive in the JR column (§4.6), and resonance amplifies beat-locked modulation more strongly than the mean rate (§4.7). Finite-amplitude QIF simulations then show how a beat-locked component coexists with gamma activity at the spike level. The laminar two-band column extends the analysis to carriers of 25–80 Hz and drives up to 3 mV (App. E).

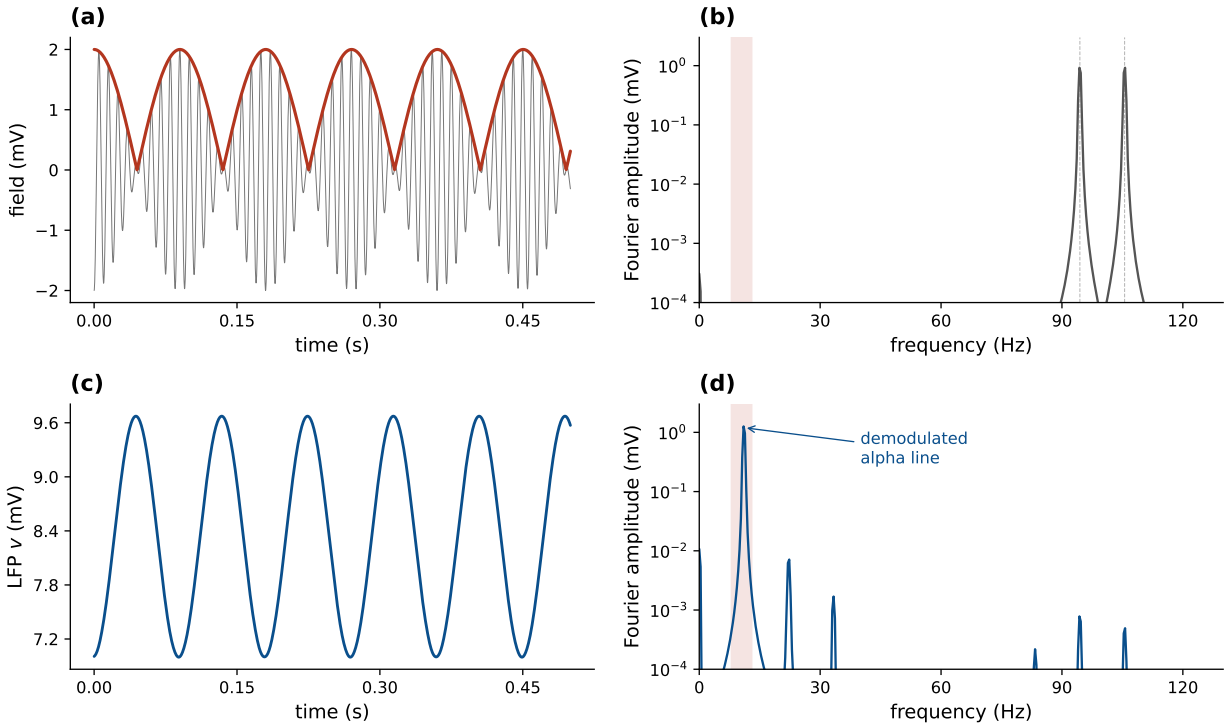


Figure 3: An alpha rhythm is synthesized from a two-tone TI input (Jansen–Rit below Hopf). The field is the physical TI waveform, two equal tones at $f_{1,2} = f_c \mp \Delta f/2$, not the amplitude-modulated surrogate. *(a)* The field $s(t)$ and its envelope $A(t) = 2\varepsilon|\cos(\Omega t/2)|$—a rectified cosine, not a sinusoid; it is A^2 that is sinusoidal at Ω (§2.2). *(b)* The input amplitude spectrum has lines only at f_1 and f_2 (94.45 and 105.55 Hz, $\Delta f = 11.1$ Hz) and no third line at the carrier; in the alpha band (shaded) it is 1.6×10^{-8} of the tone amplitude, finite-window leakage rather than physical power. *(c)* The output $v = y_1 - y_2$ oscillates at the envelope frequency. *(d)* The output spectrum has a strong peak at the beat, 11.1 Hz, synthesized by the sigmoid and absent from the input.

4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

Figure 2 orients the system: a supercritical Hopf at $p \approx 315$ with $f_0 \approx 11.1$ Hz, and a sigmoid whose curvature peaks well off the inflection, so the closed-loop operating points sit where σ'' is large. An input with no spectral power in the alpha band, with lines only at the two tones $f_c \mp \Delta f/2$, produces an output oscillation at the 11.1 Hz envelope frequency and an alpha peak in the output spectrum that is absent from the input (Fig. 3).

4.2 Near the Hopf, the demodulated beat drives a sharp resonance

The resonance is quantified in Fig. 4. In the fixed-point regime ($p > 315$, a stable focus) the response is approximately Lorentzian, peaking near f_0, and its height grows monotonically as the Hopf is approached ($0.23 \to 0.43 \to 0.70$ mV; Fig. 4), i.e. gain $\propto 1/\gamma$. This is forced gain: with no drive the focus is silent at Ω, so all power at Ω is synthesized by the demodulating nonlinearity and amplified by the recurrent loop. Above the Hopf ($p < 315$) the column free-runs at f_0. A lock-in at Ω can then contain both the forced component and a contribution from the autonomous cycle. We characterize this regime with the dominant output frequency f_{out} (Fig. S4), which identifies a frequency-dominance region but does not establish phase capture.

4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 5). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above

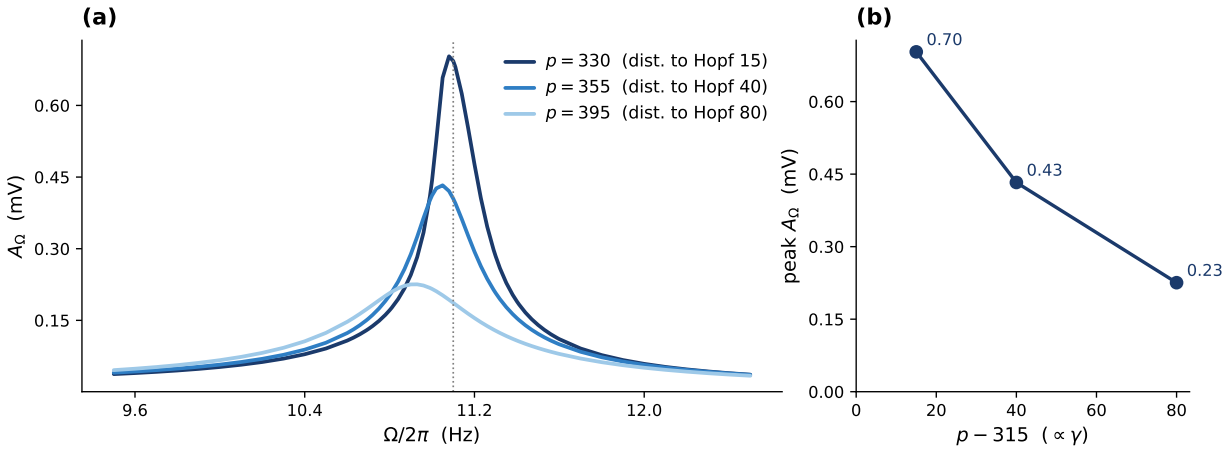


Figure 4: The demodulated beat drives a resonance that sharpens toward the Hopf (Jansen–Rit). Fixed-point regime ($p > 315$, stable focus): lock-in response at Ω vs. envelope frequency ($\varepsilon = 0.3$ mV). *(a)* The demodulated drive forces an approximately Lorentzian resonance near $f_0 \approx 11.1$ Hz. The measured finite-amplitude peak shifts from 11.08 to 10.93 Hz across the curves as the operating state changes; Eq. (20) describes its near-Hopf scaling, not its exact location. The response carries no power at Ω in the absence of drive, so this is forced gain. *(b)* The peak height grows as the Hopf is approached ($p \to 315$, distance to Hopf $\to 0$). Three points cannot establish a power law; the $1/\gamma$ scaling is quantified over a decade, together with its saturation, in Fig. 9(c). The above-Hopf regime, with autonomous alpha activity, is characterized by output frequency in Fig. S4.

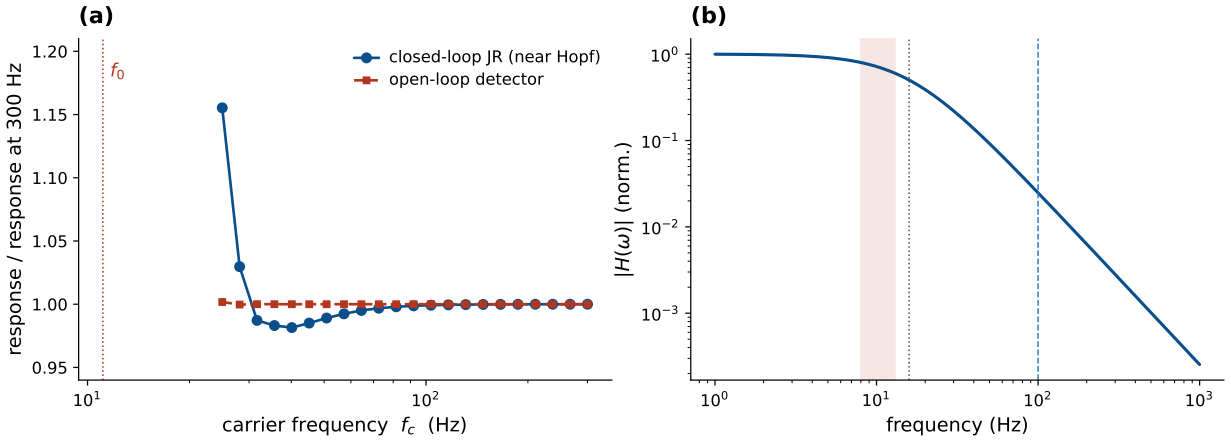


Figure 5: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is nearly flat across carrier frequency $f_c = 25$–300 Hz for the open-loop detector and for the closed-loop JR near the Hopf, with deviations as f_c approaches the natural band. Each curve is normalized by its own response at 300 Hz: 0.00670 s^{-1} open loop and 0.512 mV closed loop. Their ratio is a rate-to-voltage transfer of 76.4 mV s. The open-loop carrier independence follows to leading order from Eq. (10). *(b)* The second-order-synapse low-pass $|H_s(\omega)|/|H_s(0)|$ passes the envelope ($\sim$0.7) and suppresses the 100 Hz carrier ($\sim$2.5%). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

the corner (Fig. S5): passive somatic demodulation is suppressed four to six orders of magnitude at kHz in the 16 ms access model, whereas the assumed 0.2 ms response retains appreciable carrier access.

We also drive the full recurrent column with a kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant τ) for each coupling route before it reaches the sigmoid. The column demodulates and resonates at the recovered alpha in the fast-access case ($\tau = 0.2$ ms), as Eq. (22) predicts (Fig. S9a). Driving the fast-element route with two *genuine*

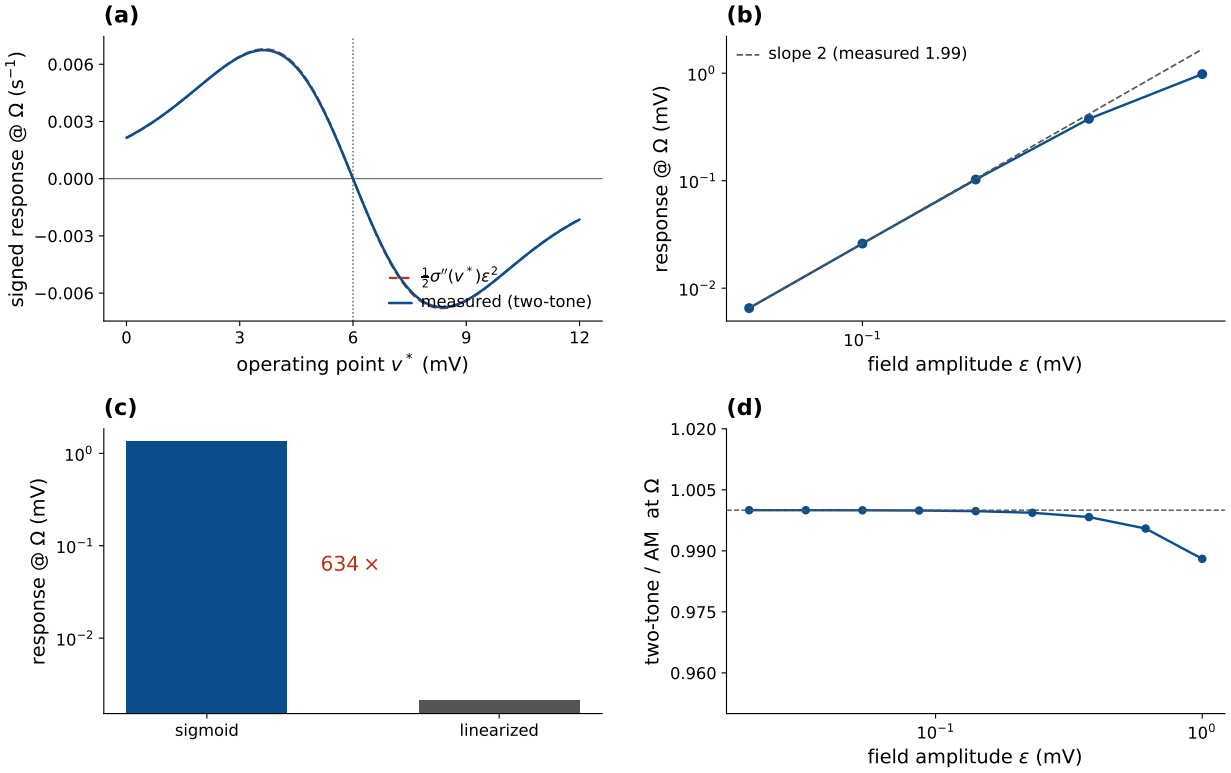


Figure 6: The curvature law, the square law, and the surrogate control (Jansen–Rit, genuine two tones). *(a)* The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2$ across the operating range (maximum deviation 0.9%), vanishing at the inflection v_0 and reversing sign across it. The mechanism is sign-agnostic: σ'' sets sign and phase while $|\sigma''|$ sets gain, so the resonance picture is unchanged up to an overall sign in the $\sigma'' > 0$ lobe, which this sweep spans. The sign flip is itself a prediction (§4.9). *(b)* The closed-loop response scales as ε^2, measured slope 1.99. *(c)* Replacing the field-receiving sigmoid by its tangent collapses the response 634×. *(d)* Surrogate control: the two-tone/AM ratio at Ω approaches 1 as $\varepsilon \to 0$ (1.0000 at 0.02 mV, 0.988 at 1 mV) for the open-loop detector at $v^* = 8$ mV. This checks the leading local source; it does not validate the recurrent AM-surrogate sweeps of Fig. 9 near the Hopf.

tones at $f_{1,2} = 1994.54$ and 2005.46 Hz, centered near 2 kHz and beating at 10.91 Hz, the finite-window input spectrum has a maximum amplitude of 2.2×10^{-13} mV in the 8–14 Hz band, while the output carries a line at the beat (Fig. S6), with the attenuation set by the analytic $|H_m|^2 = 0.137$ at 2 kHz for $\tau = 0.2$ ms. The two-tone result therefore quantifies the fast-access hypothesis under the specified response time and coupling. Limited sensitivity to unmodulated kHz fields has been reported in hippocampal network oscillations [38].

4.4 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 6)—demodulation is the sigmoid-curvature term of Eq. (10), and the closed-loop operating points sit where σ'' is large and negative.

Two controls test the proposed source of the beat component (Fig. 6b,c): the response scales as ε^2 at small drive (measured log–log slope 1.99 against the predicted 2), and replacing the field-receiving sigmoid by its linearization collapses the response by 634× (1.34 → 0.0021 mV)—the nonlinearity is the detector, and its curvature is the gain. A third panel checks the surrogate at one operating point: for the open-loop detector at $v^* = 8$ mV the two-tone and AM responses at Ω agree to within 10^{-4} at $\varepsilon = 0.02$ mV and to 1.2% at 1 mV (Fig. 6d). On that basis the dense perturbative sweeps below, which read only the Ω component, use the surrogate; the comparison does not test the recurrent response near the Hopf, where the two waveforms' different DC terms

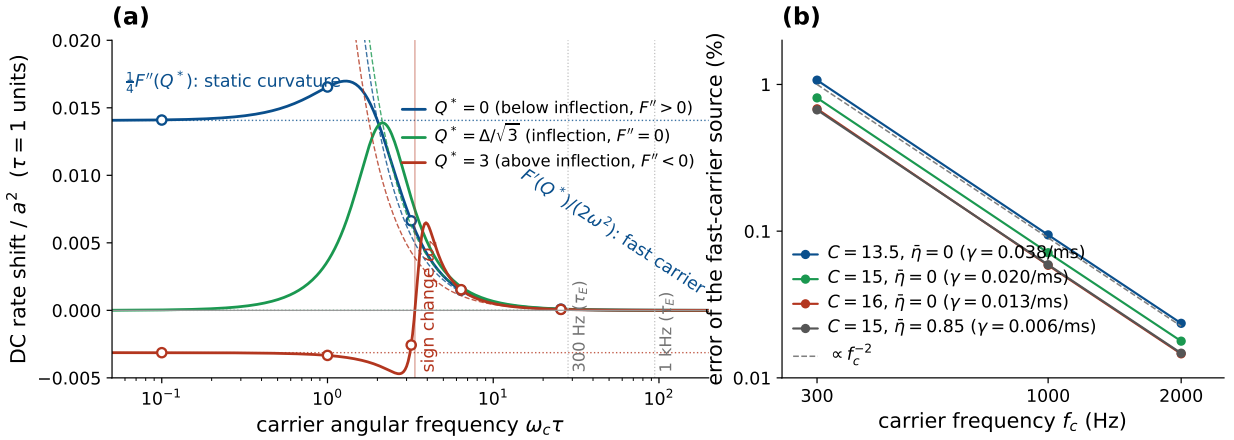


Figure 7: Static curvature at slow carriers, slope at fast ones. *(a)* Second-order DC rate shift of an uncoupled MPR population ($\Delta = 1$, $\tau = 1$) under a current carrier, Eq. (13), for operating points below, at and above the inflection of $F(Q)$. Dotted: the static limit $\frac{1}{4}F''$; dashed: the fast-carrier limit $F'/(2\omega^2)$; circles: RK4 integrations at $a = 0.05$ averaged over whole carrier periods (relative error below 10^{-3}). Above the inflection the sign changes at ω_z. The top axis converts $\omega\tau$ to a carrier frequency for $\tau = \tau_E = 15$ ms. *(b)* PING of Eq. (27): relative error of the fast-carrier difference-frequency source against the exact second-order response at four stable operating points (γ the damping of the gamma mode), two tones centered at 300, 1000 and 2000 Hz.

could shift the operating state.

4.5 Exact mean-field confirmation: the NMM2 PING gamma generator

The JR column and the LaNMM both use the heuristic firing-rate sigmoid (NMM1). As an independent test we move to the exact next-generation mean field (NMM2, Eq. (4)), where the demodulating nonlinearity is the *derived* quadratic v^2 term rather than a postulated curve. We drive the two-population E–I PING motif of Eq. (27) (Methods), which loses stability through a supercritical Hopf at $\bar{\eta}_H = 1.110$, $f_0 = 54.0$ Hz, with the two-tone field entering the excitatory current, so the carrier is rectified by the exact v_E^2 term.

Driving with a two-tone TI field (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) and locking in the excitatory rate r_E at Δf reproduces the generation and amplification of the beat (Figs. 8, S8). Below the Hopf the response is a forced resonance at the gamma frequency whose height grows as the bifurcation is approached, following the $1/\gamma$ gain of Eq. (20); above it the demodulated drive acts on an autonomous gamma rhythm. The two-dimensional map traces a ridge at $\Delta f \approx f_0$ that intensifies through the Hopf. Compared with the JR sweep shown here, the NMM2 resonant frequency shifts appreciably with the operating point, producing a diagonal ridge [30]. That the beat is generated and resonantly amplified with the nonlinearity *derived* from QIF spiking dynamics, rather than assumed, shows the receiver behavior is not an artifact of the heuristic sigmoid. The detector coefficient, however, is not the sigmoid's: at this 300 Hz carrier the population is on the fast side of its crossover, and the generated source is the v_E^2 mixing of the carrier-frequency voltage excursion, whose fast-carrier approximation predicts the second-order response to within 1.1% at the four tested stable points (§2.3, Fig. 7).

4.6 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input p, but p also shifts the operating point, entangling detection (σ'') with amplification. To isolate amplification, we vary the JR global connectivity C (coupling strength J) and solve analytically for the drive $p(C)$ that *pins* the field-free operating point v^*, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\text{open}} \simeq \frac{1}{2}|\sigma''(v^*)|\varepsilon^2 m$ are held constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$)

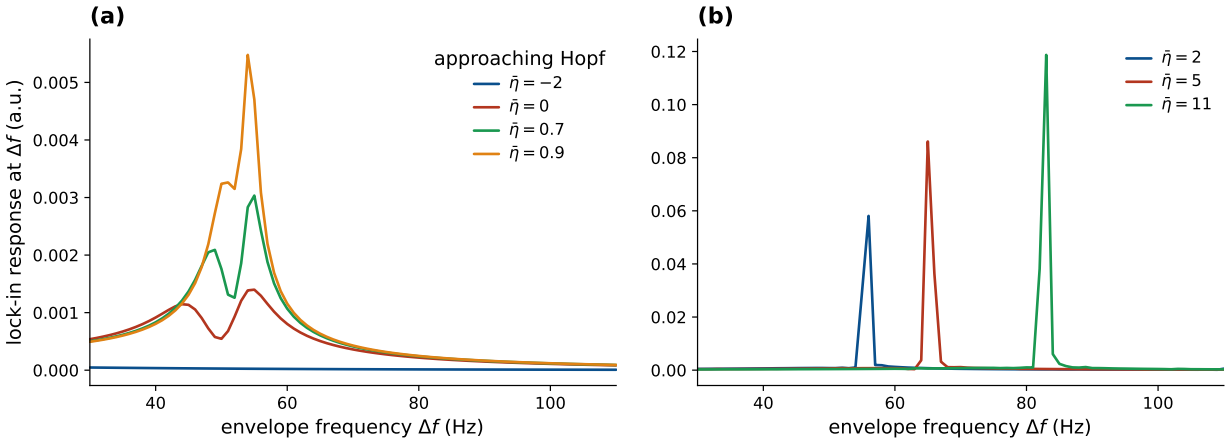


Figure 8: NMM2 PING gamma resonance (exact mean field, genuine two tones). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_H = 1.110$) forced resonance at the gamma frequency ($f_0 \approx 54$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_H$) response in the autonomous-gamma regime, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity, with no sigmoid anywhere in the model. Matched AM and two-tone drives agree at Ω to 0.07% well below the Hopf and to 4% at $\bar{\eta} = 0.9$, consistent with the JR surrogate control (Fig. 6d).

the damping $\gamma \to 0$, the demodulated response grows $\sim 30\times$ and the envelope resonance sharpens (Fig. 9a,b), a network effect since the local detector curvature is fixed while the recurrent transfer changes. Its modal frequency and input–output projections vary with C as well as its damping. The rate-to-voltage gain $A_\Omega / A_{\text{open}}$ (mV s) tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input p rather than the coupling C gives a similar leading gain–γ trend. Both paths approach a weakly damped mode; their finite-distance gains need not coincide because the modal projections can differ.

The direct low-frequency control tests whether this amplification favors TI. For the AM surrogate with amplitude ε_{TI} and modulation depth m, and a tACS input of amplitude $\varepsilon_{\text{tACS}}$, the ratio of voltage lock-in amplitudes in linear response is

$$\mathcal{R}_{\text{lin}} = \frac{|\sigma''(v^*)|\, \varepsilon_{\text{TI}}^2 m}{2\sigma'(v^*)\, \varepsilon_{\text{tACS}}}.$$

The common network transfer cancels, so the ratio has no dependence on γ when both responses are linear about the same state. At the amplitudes used in Figs. 9 and S3, $\mathcal{R}_{\text{lin}} = 0.143$. The cached sweeps give approximately 0.14 far from the Hopf, rising to 0.24 at $\gamma = 1.07$ s^{-1} and 0.47 at 0.12 s^{-1}. These are response ratios for the stated unequal input amplitudes, not an equal-power comparison. In controls at three couplings, matching the tACS input to TI's leading beat source gives TI/tACS output ratios of 0.99, 0.94, and 0.92. Increasing that direct input by a factor of 6.99 increases its response by 6.97, 3.95, and 1.96, respectively. These controls support earlier saturation of the larger response, accompanied by stimulation-induced mean shifts that field-free pinning does not cancel. The linear cancellation therefore does not extend to arbitrary finite-amplitude responses near criticality.

All these sweeps start from field-free stable foci ($\gamma > 0$), approaching the supercritical Hopf from below: the $1/\gamma$ growth is the forced-resonance susceptibility of a damped network, not entrainment of an autonomous cycle. On the other side of the bifurcation the column oscillates on its own, and the demodulated drive can phase-lock that limit cycle within an Arnold tongue. The maps here measure frequency dominance (Fig. S4 and the LaNMM and NMM2 maps), whose band grows with drive. Phase tracking would be needed to establish capture. The amplitude of an autonomous

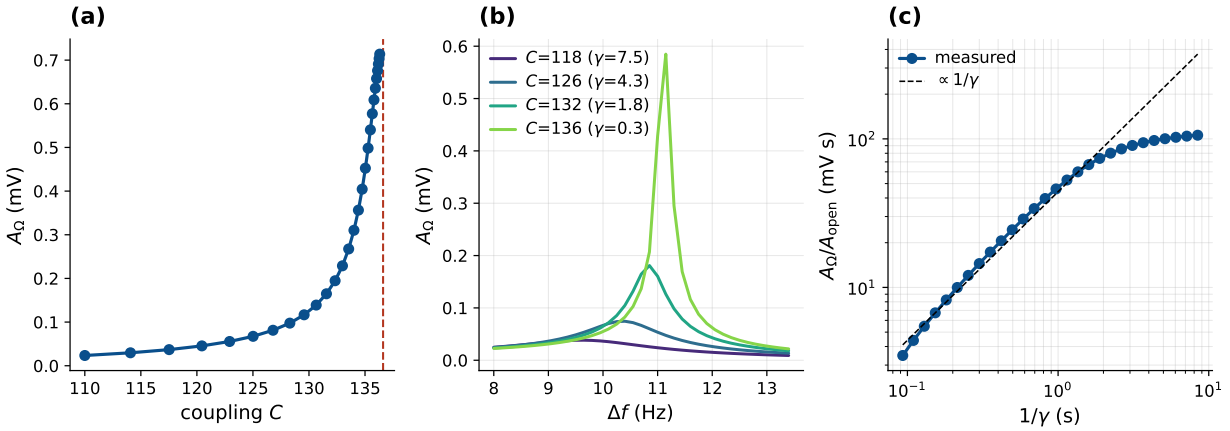


Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit). The field-free operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling $C\,(= J)$ approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain $A_\Omega/A_{\mathrm{open}}$, in mV s, follows $1/\gamma$ (dashed) and saturates near the bifurcation.

rhythm and the susceptibility of a stable focus are different observables; the $1/\gamma$ law pertains to the latter. The following rate measurements compare beat-locked modulation with the nonzero change in mean firing rate (§4.7).

The next-generation theory gives the same result without pinning. In the NMM2 PING the coupling C is the *literal* mean inter-neuron (QIF) synaptic weight J. The leading fast-carrier source provides the normalization for this unpinned sweep (§2.3); finite-carrier source corrections over the full coupling range have not been bounded. Sweeping J at fixed background drive $\bar{\eta}$ toward the gamma Hopf J^*, the demodulated peak of r_E grows as $1/\gamma$ before departing from this scaling near criticality (Fig. S10): the heuristic JR result, reproduced with the coupling *and* the rectifier both derived rather than assumed, and a direct counterpart of the coupling-controlled weak-field sensitivity of QIF networks [30]. Here γ describes the field-free equilibrium. Although a DC-only averaged calculation predicts an unstable endpoint, the full periodically forced solution remains attracting: its largest perturbation multiplier is 0.034 over the 2 s forcing period. Increasing settling from 0.6 to 3 s changes the endpoint lock-in by 0.8%. The final points therefore describe a stable nonlinear driven response rather than linear gain about the DC-shifted equilibrium. The two columns exhibit the same gain scaling in their two native bands, alpha (JR) and gamma (NMM2): the single-cell nonlinearity detects the envelope (coarse-grained into σ'', or the exact v^2), while the gain is a recurrent amplification set by proximity to the Hopf and adjustable through the coupling.

4.7 The resonance amplifies the envelope-locked modulation more than the mean rate

The demodulator feeds the network a constant term and a line at the beat Ω (Eq. (9)), and the transfer $G(\Omega)$ of Eq. (20) treats them differently: $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $G(0) \propto 1/(\omega_0^2 + \gamma^2)$ stays finite. Pinning v^* to hold the local detector curvature fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, we read both quantities from the same observable, the pyramidal rate $r_P = \sigma(v+s)$ with the field in its argument. The beat-locked modulation of r_P at $\Delta f = f_0$ grows 13.5× (26 to 346 mHz), tracking $1/\gamma$ before saturating near criticality; the shift of its mean grows too, 5.8× (−1.3 to −7.7 mHz), through rectification of the growing oscillation, but stays below 0.2% of the 3.9 Hz baseline (Fig. S11). Modulation exceeds mean shift by 19× far from the Hopf and by 45–66× near it. The network amplifies the envelope-locked modulation far more than the mean rate, so a small mean-rate change is the expected signature even where TI is acting, the mesoscopic counterpart of the observation that TI alters spike timing more than firing rate in the primate brain [19]. The mean shift is small

but nonzero, so the comparison must use commensurate measures, as here.

4.8 QIF responses distinguish beat modulation from field-induced gamma

The QIF network connects the population description to individual spikes [28, 25, 30]. With 8000 neurons per population and the same parameters as the PING mean field, it distinguishes a forced beat response from a stimulation-induced change of dynamical regime (Figs. 10, S13).

At $\bar{\eta} = 0.85$ and tone amplitude $A_f = 10$, the spiking network has a beat depth of 0.093 at 55 Hz and a mean-rate increase of 2.8%: modulation exceeds the fractional mean-rate change by a factor of 3.4. The mean field gives a larger depth, 0.128. Raising the spike cutoff from $V_{\text{peak}} = 100$ to 200 and 400 gives depths 0.127 and 0.139, while the mean-rate increase stays at 2.8–2.9%. At the original cutoff, doubling the population gives 0.083 and halving the time step gives 0.093. The response is therefore sensitive to the finite spike cutoff. Halving the step at cutoffs 200 and 400 gives depths 0.121 and 0.141, respectively: cutoff dependence persists, and quantitative convergence remains unestablished at this population size. At the plotted cutoff, the pooled spike-phase vector strength is 0.046; median per-neuron phase consistency remains negative in field-off and field-on windows at all three cutoffs. At $\bar{\eta} = 1.5$, the corresponding 42 Hz depth is smaller, 0.023, against a 1.6% mean-rate increase, and the existing gamma dominates the raster.

At the larger amplitude $A_f = 35$ and $\bar{\eta} = 0.85$, stimulation produces a dominant line near 52 Hz, with modulation depth about 0.70, and a smaller component at the 55 Hz beat, with depth 0.29. The second-order carrier-averaged system predicts an excitatory drive shift $\delta\eta_E \approx 1.65$, beyond the E-only Hopf threshold $\delta\eta_E^* \approx 0.286$, with a leading eigenvalue whose real part is $+0.032$ ms^{-1}. Direct mean-field integration reproduces the off-beat line, supporting the interpretation of panel (b) as field-induced gamma with a smaller forced component.

The equal-RMS unmodulated carrier provides a control for this distinction in the mean field (Fig. S13c,d). It produces a rectified E-drive shift within 3% of TI's and a similar mean-rate increase. At $A_f = 35$, both waveforms induce gamma near 51 Hz, whereas TI also produces the 55 Hz beat component. Increasing the measurement window from 0.8 to 3.2 s reduces the carrier's apparent 55 Hz depth from 0.011 to 9.2×10^{-5}, consistent with spectral leakage, while TI's beat depth remains 0.30–0.31. At $A_f = 10$, neither induces gamma, and TI alone produces a resolved beat response. The control separates induction of an oscillation from its beat-frequency response; it has not been run in the spiking network.

At $\bar{\eta} = 1.5$, gamma activity is already present without stimulation. With the field on, the rhythm near 53–54 Hz remains about four times larger than the 42 Hz beat-locked component. Gamma therefore dominates the raster in panel (d); the coexistence of the rhythms produces slower variations in burst height. Folding the rate on the beat phase suppresses the non-beat-locked gamma and exposes the smaller component. The inset shows that phase average, not the full rate variation.

The population spectrum also contains a sum sideband near 95 Hz and a smaller difference component near 11 Hz (Fig. S12b). These lines establish nonlinear mixing, without uniquely identifying gamma-amplitude gating or phase modulation. The field-free gamma spectra of the spiking network and exact mean field agree in frequency (Fig. S12a).

At $A_f = 35$, the mean rate rises by about 8–11% across the two conditions, while the beat modulation depth, $A_\Omega/\langle r_E \rangle$, rises from below 0.003 to 0.29 ($\bar{\eta} = 0.85$) and 0.23 ($\bar{\eta} = 1.5$) (Fig. S12c), measured over 44 and 34 whole beat cycles after 800 ms of settling (Methods). Timing modulation therefore exceeds the relative mean-rate change by a factor of about three at the population level. Per neuron the locking is weak: the pairwise phase consistency of single-neuron spike phases at Δf rises from its field-free control (median -0.02) to a median of 0.003–0.005, and the pooled vector strength from below 0.02 to 0.12–0.13. The population's beat-locked component is built from many weakly biased cells. This gives a qualitative counterpart of the timing-versus-rate distinction in

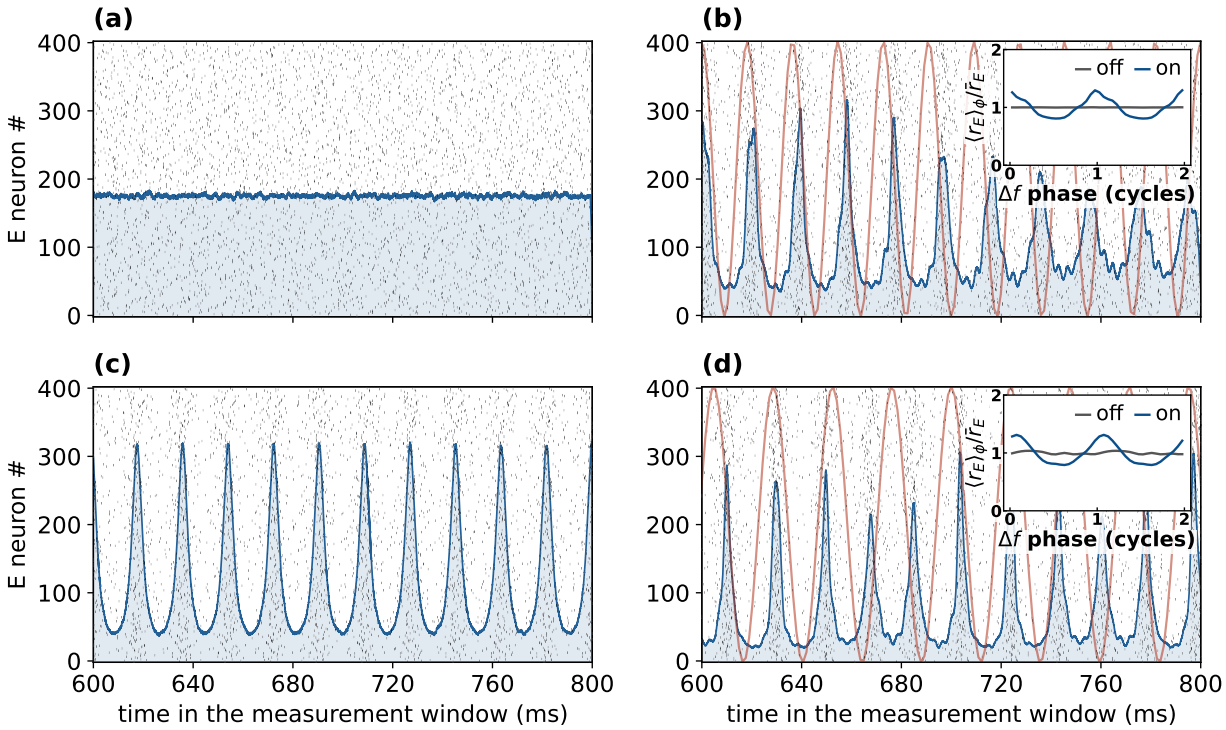


Figure 10: Beat-locked and gamma activity coexist in the QIF network. $N = 8000$ excitatory and inhibitory quadratic integrate-and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (27)). These are finite-amplitude examples ($A_f = 35$ per tone, carrier $f_c = 300$ Hz). Black ticks are E spikes (400 of 8000), blue is the smoothed population rate on a separate scale, and red is the TI envelope in arbitrary units. *(a,b)* Field-free stable state ($\bar{\eta} = 0.85$), TI off then on at $\Delta f = 55$ Hz. The field-on response is dominated by field-induced gamma near 52 Hz (depth 0.70), with a smaller beat component (depth 0.29). *(c,d)* Field-free oscillatory state ($\bar{\eta} = 1.5$), TI off then on at $\Delta f = 42$ Hz. Gamma near 53–54 Hz dominates the raster, with depth about four times the beat component's 0.23; their coexistence gives slow variations in burst height. Depth is line amplitude divided by mean rate. The final 200 ms of the common measurement interval are shown. *Insets* in *(b,d)* fold the rate on beat phase over 44 and 34 cycles (off gray, on blue), normalized to each condition's mean. This averaging suppresses non-beat-locked gamma and displays the smaller beat component on a common 0–2 scale. The fold alone does not establish gamma-amplitude gating or phase capture.

§4.7 and in primate recordings [19]; the present amplitudes do not establish the size of that effect under weak physiological stimulation.

4.9 Falsifiable predictions and tests

The mechanism makes four predictions, each with an experimental handle.

1. **Carrier dependence separates access from mixing.** In the static limit and in the fast-carrier QIF asymptote, the leading beat is independent of f_c at fixed local polarization, provided the carrier does not excite another network mode. At fixed applied field, access restores frequency dependence (Eq. (22)); detector dynamics can add it near the crossover. *Test:* measure or estimate local polarization during a carrier sweep and compare the response before and after correcting for access. The remaining frequency dependence constrains the mixing mechanism. Its cutoff is to be measured, not assigned to a particular compartment in advance.

2. **State- and carrier-dependent response.** The generated beat depends on the operating point through the detector coefficient, and the two modeled detectors disagree on how: the memoryless sigmoid follows $\sigma''(v^*)$ and reverses sign across the inflection, where PEIX (Eq. (24)) peaks; the dynamic mean field at a fast carrier mixes in the voltage dynamics and is read through

the rate and network susceptibility; its DC rectification follows F', without the same curvature sign rule (§2.3). *Test:* vary tonic drive and carrier frequency while measuring both amplitude and phase of the beat response. Compare these with the predicted mixing coefficient after accounting for changes in network susceptibility; a response-amplitude change alone does not isolate the detector.

3. **Transduction class.** A smooth memoryless response with nonzero curvature transduces A^2 at weak input, whereas an ideal envelope follower transduces A (§2.4), and the two are separable at a single matched target. The former gives $Y_\Omega \propto \varepsilon^2$ with a second-harmonic ratio vanishing as $\varepsilon \to 0$; the latter gives $Y_\Omega \propto \varepsilon$ with the ratio pinned at $1/5$. *Test:* sweep field amplitude off resonance (so that both lines sit on the flat part of G) and read the log–log slope together with $Y_{2\Omega}/Y_\Omega$; on resonance the network suppresses the second harmonic by $\sim 3\mathcal{Q}$ and the test must be corrected by an independently measured $G(2\Omega)/G(\Omega)$.

4. **Frequency selectivity and state-dependence.** The response is largest when Δf matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation. Changes in operating state can therefore alter both the generated drive and the recurrent gain [23]. *Test:* sweep Δf through a region's intrinsic band across brain states and track the resonant peak and its width.

Table 3 compares the predictions and related model consequences with existing evidence. Two are qualitatively consistent with published observations: TI alters spike timing more than firing rate in vivo [19] (our $G(\omega_0) \gg G(0)$ asymmetry, §4.7–4.8), and most pyramidal TI responses vanish when synaptic transmission is blocked [15] (a network read-out, which tests network involvement rather than excluding a surviving cellular detector). A quantitative reanalysis of these datasets against the model is a natural next step, and §5 specifies the amplitude-exponent fit it would require.

Prediction (model basis)	**Existing evidence**	**Proposed test**
Transduction class: $Y_\Omega \propto \varepsilon^2$ and vanishing $Y_{2\Omega}/Y_\Omega$ (square law) vs. $\propto \varepsilon$ and $1/5$ (envelope following)	— (untested; existing comparisons match extracellular field, not ε)	Amplitude sweep off resonance; transfer-corrected harmonic ratio
Timing-dominated response: the envelope-locked modulation of the rate grows toward the Hopf faster than its mean ($G(\omega_0) \gg G(0)$)	*Qualitatively consistent*: TI alters spike timing more than firing rate, primate [19]	Lock-in of spike timing at Δf vs. the mean-rate change, in the same effect measure
Network read-out (population nonlinearity + synaptic recurrence)	*Qualitatively consistent*: most pyramidal TI responses vanish under synaptic block [15]	Synaptic block should reduce recurrent amplification; cellular mixing may persist
Carrier independence at fixed polarization (no ω_c in A_Ω)	TI thresholds inherit the carrier's f_c-dependence [20, 21]; bare kHz weakly effective [38]	Fix in-zone polarization, sweep $f_c \to$ flat envelope response in the stated limits; fit access and residual detector dependence
State- and carrier-dependent response (the detector coefficient is model-dependent)	Cell-type-specific TI responses (PV vs. pyramidal) [15]	Shift v^* by state at fixed beat; sweep f_c at fixed beat
Frequency selectivity, sharpened $\sim 1/\gamma$ near criticality	Quantitative gain–bandwidth relation not tested here in biological data	Sweep Δf through a region's band across brain states

Table 3: The mechanism's predictions confronted with existing in-vivo / in-vitro evidence. Entries cite documented qualitative findings; quantitative reanalysis is left to future work.

5 Discussion

Fast axonal access provides a candidate first stage of the TI mechanismAxonal ion-channel rectification is already demonstrated in single-neuron TI models [2], with axonal excitation and block also found in cortical-neuron models [8]. We represent candidate fast axonal access by a short effective membrane time in the population model. Cohen et al.'s periaxonal circuit [6] establishes distinct voltage dynamics across myelin and the axolemma, motivating an axonal access response separate from the somatic one. Nodes and the AIS are candidate sites where polarization can encounter voltage-gated nonlinearities[7, 2, 8]. The proposed connection is from this ; the AIS has a high sodium-channel density [7]. The population reduction connects local polarization to cellular mixing and then to recurrent amplification. Luff et al.'s subthreshold mixing [9] supplies evidence for the detector; it does not localize that detector to the AIS or the periaxonal circuit.

At applied extracellular fields of order 1 V/m, voltage-gated channels respond through membrane polarization, not through direct action of the field on the protein: the structural effects reviewed by Cifra et al. [39] are reported at fields from 10 kV/m to GV/m, with thresholds that depend on the protocol and molecular system. That review places fields below 0.1 kV/m outside its scope, where it expects direct effects to be negligible against thermal noise. Presynaptic terminals can polarize more strongly than somata [40, 41, 42], but small size alone does not imply fast charging: for an isopotential membrane, the product of specific resistance and capacitance is independent of area. Conductance, geometry, and the mode of stimulation determine the response.

The time scales in Table 2 refer to different measurements and models. Cohen's approximately 1 ms minimum axonal decay constrains an injected-current response. Luff et al.'s pulse-width-modulated TI (PWM-TI) model uses a 1 ms passive response within a 1–100 ms parameter sweep [43]. Separately, Semenov et al. [22] measured field-induced membrane charging near 0.7 μs in isolated neurons under strong, microsecond field pulses. This is a different response mode from intracellular current charging or periaxonal conduction. Together these studies motivate examining a shorter effective access time, while leaving its value and field coupling preparation-dependent. A 10–100-fold reduction in a QIF membrane time is therefore a sensitivity hypothesis. Its population-level consequence can be studied through the existing slow-synapse limit, with each physical filter included once; a new periaxonal circuit is not needed for that limiting argument.

Once the field reaches a nonlinearity, weak-field TI admits an effective description in which cellular contributions generate a quadratic mixing response and recurrent dynamics amplify it. For a smooth response with a nonzero second-order term, microscopic contributions combine in the mixing coefficient; identifying each channel is unnecessary for this leading-order account. QIF provides a derived realization. This connects cellular mixing [2, 9] with the network dependence seen in slices and simulations [15, 16, 17]. Cellular mixing and circuit amplification are therefore not competing explanations.

Small amplitude and fast relaxation justify different reductions. Weak input permits a second-order expansion of the dynamic response. Fast membrane relaxation gives the NMM2 slow-synapse limit, a static rate curve with dynamic synapses; if relaxation is also fast relative to the carrier, its curvature gives the leading beat. Otherwise the mixing coefficient retains frequency dependence. The curve's slope converts a slow input into a rate change and sets local recurrent gain. The generic quadratic structure survives this distinction, while its magnitude, phase, and frequency dependence remain model-specific.

A spatial TI dose map depends on this physiology. The usual envelope-modulation measure $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$ assumes that neural response is determined by the envelope excursion. In the local square-law regime, the beat instead depends on $\varepsilon_1\varepsilon_2$, and the carrier average also contains a DC term proportional to $\varepsilon_1^2 + \varepsilon_2^2$. Orientation, access, operating point, and network susceptibility further shape the response. For a smooth memoryless transfer, carrier averaging is even in the envelope amplitude, giving quadratic transduction when the second-order coefficient

is nonzero. Dynamic nonlinearities can also mix at second order, with frequency-dependent coefficients. A sharp rectifier can instead approximate envelope following over a finite amplitude range; smoothing or heterogeneity can restore quadratic behavior at smaller inputs. The amplitude and harmonic tests in §2.4 distinguish these regimes.

Stimulation intensity can also change the mechanism. At high intensities, TI and unmodulated kHz fields can have similar excitation thresholds, with modulation patterning firing rather than lowering the threshold [20, 22]; some peripheral responses do not require envelope extraction [21]. The weak-input calculation here concerns modulation of ongoing activity through mixing and amplification [19, 44]. Direct carrier-evoked spiking and conduction block are outside its scope. The finite-amplitude QIF examples illustrate a related distinction: rectification can shift a stable population across its Hopf and induce gamma activity alongside the beat-locked component. The driven component must be separated from that change in regime. The boundary depends on the initial state's distance from the bifurcation as well as on field amplitude. These examples and the laminar extension describe behavior beyond the weak-input expansion; they do not establish physiological thresholds.

The recurrent amplifier is shared with direct low-frequency stimulation. In our coupling sweep, holding the JR field-free operating point fixed holds the local curvature fixed while the network gain changes. A direct in-band drive shows the same $1/\gamma$ growth and resonance sharpening (Fig. S3), consistent with the known resonant susceptibility of recurrent excitatory–inhibitory networks [14, 30]. Near the Hopf, this is temporal critical slowing in a mean-field model, not evidence of a diverging spatial correlation length. Related models already reproduce tACS entrainment and AM-driven phase locking [45, 46, 47]. The separation here is between the drive generated by mixing and the response to that drive. When TI and tACS are matched for effective in-band input, observable, and operating state, they share the linear amplifier. Within linear response, resonance alone therefore gives TI no efficiency advantage. At finite amplitude the response ratio can change near the Hopf, as the coupling sweep demonstrates (§4.6); its value then depends on drive amplitude and the driven state as well as on field-free damping.

Absolute response estimates remain sensitive to access and transduction. Measured static field-to-voltage couplings are 0.12–0.21 mV per V/m in hippocampus [48, 49] and reach about 0.5 mV per V/m in cortical L5/6 pyramidal cells [50]; human cortical fields near 0.8 V/m have been reported at 2 mA [51, 52]. These static couplings do not fix the coupling at kHz. For equal TI carrier polarizations ε, the JR beat drive is $\sigma''\varepsilon^2/2$, whereas a direct drive of the same peak polarization gives $2\sigma'\varepsilon$. Their leading amplitude ratio is $\eta_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$: the relative efficiency decreases as the input is weakened in this perturbative regime. A larger recurrent gain amplifies both drives.

For scale, a reverberation time $\tau_{\mathrm{rev}} = 1/\gamma$ gives a near-Hopf quality factor $\mathcal{Q} \simeq \pi f_0 \tau_{\mathrm{rev}}$ and power bandwidth approximately $1/(\pi\tau_{\mathrm{rev}})$. Population relaxation over a few hundred milliseconds [53] would give a quality factor of order ten at 10 Hz and a bandwidth of order 1 Hz. This is an order-of-magnitude comparison: the measured relaxation is not specific to the alpha mode. Narrow amplification also makes detuning consequential.

Primate measurements illustrate both the relevance and the limits of an efficiency comparison. Vieira et al. [19] used modulation depths of 0.5–0.62 V/m and reported TI effects at 0.34 times those of conventional tACS (95% confidence interval 0.04–0.65). Their within-electrode AM comparison compensated for frequency-dependent current shunting and matched extracellular modulation depth near the recorded cells. A favorable fast-access estimate with $\varepsilon \approx 0.1$ mV gives only about 0.8% for the JR detector, below the reported ratio. This discrepancy is not a calibrated rejection of that detector: the comparison involved twenty neurons, with ten responding to at least one modality and seven to both; phase-locking value (PLV) depends on baseline locking and network state; and the matched extracellular fields do not ensure matched polarization at the cellular mixer. An unspecified nonlinear relation between generated drive and PLV also prevents identification of the input-amplitude exponent. The data establish a weaker AM effect than the

matched-depth low-frequency drive, but do not isolate the transduction law.

For comparison, the matched-peak efficiency over the logistic family has a supremum of approximately 0.348 (App. D). Approaching it at weak fields requires a transfer curve narrowed to microvolt scales; the limiting detector is step-like and has a second-harmonic ratio near 0.15, distinct from both weak square-law detection and linear envelope following. This bound applies to that memoryless family and matching convention, not to the dynamic QIF detector or to all cellular mechanisms. Its proximity to an experimental ratio is not a fit.

Quantitative physiological predictions require calibration of access and mixing. Their values are not fixed by the quadratic scaling law. In comparisons of stimulus-evoked firing, Hodgkin–Huxley and Frankenhaeuser–Huxley models reproduce responses missed by the tested leaky, exponential, and adaptive-exponential integrate-and-fire models [54]. Such model dependence concerns the specific response, without invalidating the weak-input expansion. Noise, heterogeneity beyond the assumed excitability distribution, and interactions between columns can broaden or shift the deterministic resonances. Cascading ordinary synaptic filters cannot solve carrier access, because they attenuate the carrier before a subsequent nonlinearity receives it. Human results are also mixed, including reported peripheral effects without central effects [55] and null occipito-parietal alpha-power modulation [56]. Testing the proposed mechanism requires measuring access, generated beat, and network tuning in the same preparation.

6 Conclusion

TI connects field-to-membrane access, nonlinear detection, and amplification. Fast axonal compartments, including the AIS and nodes, offer a candidate access route motivated by distinct axonal voltage dynamics and periaxonal circuitryBuilding on axonal rectification in existing single-neuron TI models, we represent candidate fast access through nodes and the AIS by a short effective membrane time. Periaxonal circuitry motivates separate axonal and somatic voltage responses. Their field-to-voltage transfer under TI and the resulting response at human field strengths require calibration.

For weak perturbations of ongoing activity, smooth cellular nonlinearities with a nonzero second-order response can be represented by effective quadratic mixing. The population models provide realizations of this description: mixing generates the difference-frequency component, and recurrent dynamics amplify it near a network resonance. A large network response does not, by itself, imply efficient detection or identify the microscopic mixer.

The static and dynamic population descriptions are connected by their time scales. Fast membrane relaxation gives the NMM2 slow-synapse limit: a nonlinear static rate map coupled through dynamic synapses. Its curvature describes carrier mixing when relaxation is also fast relative to the carrier. Outside that limit, the dynamic response must be used. Fast cellular access and slow recurrent resonance can therefore coexist, with each physical filtering process represented once.

In the weak square-law regime, beat generation depends on the product of the effective carrier inputs, not only on the envelope excursion. TI and a direct low-frequency drive share the downstream amplifier when input, observable, and state are matched and both responses remain linear, so resonance alone gives TI no efficiency advantage. Finite-amplitude responses near the Hopf can depart from that ratio. Measuring local carrier polarization, the generated beat, and circuit gain separately would calibrate the stages of the proposed mechanism.

Code and data availability

Simulation code, cached results, and figure-generation scripts are provided at the paper's GitHub repository; the public archive is linked through Zenodo (doi:10.5281/zenodo.21009618). The repository `README` maps figures to their scripts and distinguishes recomputation from replotting cached data. A reconstructed producer with explicit run settings reproduces every archived spectral bin

of the two-tone kHz illustration (Fig. S6) exactly. The LaNMM setup is a drawn schematic. The LaNMM carrier comparison has both measurement-window caches and a reproducible summary script. The code is released under CC-BY-4.0.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

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A Notation and units

These neural-mass models are phenomenological: potentials are in mV, firing rates in s^{-1} (≡Hz), time constants in ms, and frequencies in Hz, unless noted; a "–" marks a dimensionless quantity. A few glyphs are reused across the four models (JR, LaNMM, NMM2/MPR, QIF) by long-standing convention; these are disambiguated by context and flagged below. In particular, A without an argument is the excitatory postsynaptic-potential (PSP) *gain* in the JR/LaNMM synapses; $A(t)$ is the instantaneous envelope of the two-tone field and A_Ω a lock-in amplitude. The applied-field modulation amplitude is ε in JR and LaNMM alike, and A_f in NMM2/QIF. These are not interchangeable units. In JR and LaNMM the field is a membrane polarization in mV entering the sigmoid argument, so ε is measured against the sigmoid voltage scale $1/\rho \approx 1.8$ mV, which is the same in both models. In NMM2/QIF the field enters the excitatory membrane current in the dimensionless MPR units of v_E, and A_f carries no volt scale at all. Likewise C is the JR connectivity constant and the NMM2 global synaptic gain $C\,(\equiv J)$.

Symbol	Meaning	Units
Firing-rate nonlinearity (sigmoid)		
$\sigma(v)$	population firing-rate transfer function	s^{-1}
$\sigma'(v)$	sigmoid slope (read-out gain)	$s^{-1}\,mV^{-1}$
$\sigma''(v)$	sigmoid curvature; the memoryless detector coefficient	$s^{-1}\,mV^{-2}$
$F(Q)$, F', F''	MPR steady-state rate curve, Eq. (5), and its derivatives	–
$C(Q,\omega)$, $K(\omega)$	MPR second-order DC coefficient at fixed current, Eq. (13), and at fixed voltage excursion, Eq. (15); d, ν the population's relaxation rate and ringing frequency	–
e_0	half the maximal firing rate ($\sigma \to 2e_0$ as $v \to \infty$)	s^{-1}
v_0	sigmoid inflection (half-maximal) potential	mV
ρ	sigmoid slope parameter	mV^{-1}
v^*	operating point (field-free fixed point of v)	mV
$v = y_1 - y_2$	net pyramidal membrane potential (LFP proxy)	mV
PEIX	Population Excitation–Inhibition Index (a σ'-based brain-state index)	–
Single-column Jansen–Rit		
A, B	excitatory, inhibitory PSP gains	mV
a, b	excitatory, inhibitory synaptic rates (inverse time constants)	s^{-1}
y_1, y_2	excitatory, inhibitory post-synaptic potentials	mV
C, $C_1 - C_4$	intracolumn connectivity constants	–
p	external input rate; JR bifurcation parameter	s^{-1}
Applied TI field and coupling		
ε	field modulation amplitude, JR and LaNMM (a membrane polarization entering the sigmoid argument)	mV
A_f, $I_E(t)$	amplitude of the field's current-like forcing $I_E(t) = P\,E(t)$ in the E voltage equation, NMM2/QIF (dimensionless MPR units of v_E)	–
m	modulation depth	–
f_c, $\omega_c = 2\pi f_c$	carrier frequency	Hz, $rad\,s^{-1}$
f_1, f_2	the two applied carrier tones ($f_c = (f_1 + f_2)/2$)	Hz
Δf, $\Omega = 2\pi\Delta f$	envelope (beat) frequency, $\Delta f = f_2 - f_1$	Hz, $rad\,s^{-1}$
$E(t)$, $s(t)$	applied field; membrane perturbation reaching the nonlinearity ($s = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}$, Eq. (8))	$V\,m^{-1}$; mV
λ, λ_0^c, λ_f^c	field-to-membrane coupling: generic, DC (door c), and carrier-effective ($\lambda_f^c = \lambda_0^c \lvert H_c \rvert$)	mV / ($V\,m^{-1}$)
$\Lambda[\hat{E}]$	direct (non-synaptic) field term in the sigmoid argument, Eq. (3); equals $s(t)$ in the JR column	mV
$H_c(\omega)$, H_{gate}	access filter of compartment c; optional channel-kinetic filter	–
$H_s(\omega)$	JR synaptic transfer function, §2.5	mV s

Symbol	Meaning	Units
$H_m(\omega),\ \tau_m$	effective membrane low-pass and its time constant	–; ms
$\tau_s,\ \tau_{\mathrm{rel}}$	synaptic and population relaxation times in the adiabatic reduction	ms
s_{syn}	filtered firing rate in Eq. (6), distinct from field polarization $s(t)$	s^{-1}
τ	effective access time used in the filter comparisons	ms
Network dynamics, response, and bifurcation		
$\mu_\pm = -\gamma \pm i\omega_0$	leading Jacobian eigenvalue pair	s^{-1}
γ	damping of the critical mode ($\propto$ distance to the Hopf)	s^{-1}
$\omega_0,\ f_0 = \omega_0/2\pi$	natural (Hopf) frequency	$\mathrm{rad\,s}^{-1}$, Hz
$\tau_{\mathrm{rev}} = 1/\gamma$	network reverberation time	s
$\mathcal{Q} = \omega_0/2\gamma$	near-Hopf resonance quality factor	–
$G(\Omega)$	response to a generated beat drive; $\propto 1/\gamma$ near resonance	output/input dependent
D	beat-source coefficient for the injection convention in Eq. (21)	model-dependent
$\chi^{(2)}(\omega_2, -\omega_1)$	effective second-order coefficient of the generated rate at fixed synaptic input, §2.2; $\sigma''(v^*)/2$ in the memoryless limit	model-dependent
A_Ω	demodulated response: lock-in amplitude at Ω of v (mV) or, in §4.7 and the QIF network, of a rate	mV or s^{-1}
Y_Ω, Y_{LF}	detector (pre-network) output at Ω and at baseband, §2.4	–
$\mathcal{M}[\cdot]$	peak-to-peak excursion of a periodic quantity, Eq. (18)	–
η_{det}	TI/tACS drive ratio of a memoryless transfer at matched peak polarization, App. D	–
δ	threshold offset of a step detector, App. D	mV
$\ell(\omega)$, ℓ_0	AC space constant and its DC value, Eq. (23)	μm
PPC	pairwise phase consistency of single-neuron spike phases at Δf [34]	–
A_{open}	open-loop rate lock-in (detector only); the JR ratio $A_\Omega/A_{\mathrm{open}}$ has units mV s	s^{-1}
I, Q	in-phase, quadrature lock-in components	readout units
C^*, J^*	coupling at the Hopf (criticality)	–
Laminar model (LaNMM)		
P_1, P_2	deep (alpha) and superficial (gamma) pyramidal populations	–
PV, SS, SST	parvalbumin, stellate, somatostatin interneurons	–
Exact mean field (NMM2 / MPR) and QIF network		
r_E, r_I	population firing rates (E, I)	s^{-1}
v_E, v_I	mean membrane potentials (E, I), in dimensionless MPR units	–
$\eta_j,\ \bar{\eta}$	per-neuron excitability; mean excitability (= common drive, bifurcation param.)	–
Δ	Lorentzian half-width of η_j (excitability heterogeneity)	–
$\tau_E, \tau_I, \tau_a, \tau_g$	E, I membrane and AMPA, GABA synaptic time constants	ms
$C\ (\equiv J)$	global synaptic coupling (inter-population gain)	–
$A_{ee}, A_{ei}, A_{ie}, A_{ii}$	synaptic weight matrix $(1, 1, 1, 2)$	–
s_E, s_I	second-order synaptic activations; $\hat{K}_a, \hat{K}_g$ their operators	s^{-1}
$V_j,\ V_{\mathrm{peak}}$	QIF neuron potential; spike/reset threshold	mV
N	neurons per population (QIF; $N = 8000$)	–
$A_\Omega/\langle r_E \rangle$	Δf modulation depth (envelope-locked timing metric)	–

Table 4: Symbols, meanings, and (nominal) units. Reused glyphs are disambiguated by context; see the head note on A and C.

B Simulated models: equations and parameters

The models are written in the NMM1 transfer-function formalism of Eq. (3) [25]: synaptic operators $\hat{L}$ transduce firing rates into PSPs, the PSPs sum with any direct field term $\Lambda[\hat{E}]$ in the sigmoid argument, and the sigmoid returns the rate. The Jansen–Rit column is given here; the laminar model is specified in [23, 31] with its field drive in Eq. (26), and the NMM2 PING motif and its QIF network in Eq. (27) and §3.4.

Jansen–Rit column

Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, external input p, and the applied-field perturbation $s(t)$ (§2.2) entering the pyramidal sigmoid,

$$\ddot{y}_0 = Aa\,\sigma(y_1 - y_2 + s(t)) - 2a\dot{y}_0 - a^2 y_0, \tag{28}$$

$$\ddot{y}_1 = Aa\,[\,p + C_2\,\sigma(C_1 y_0)\,] - 2a\dot{y}_1 - a^2 y_1, \tag{29}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{30}$$

integrated as six first-order ODEs; the LFP proxy is $v = y_1 - y_2$ [26].

Table 5: Jansen–Rit parameters [26].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

C The cubic Hopf normal form lacks a quadratic mixer

The Stuart–Landau equation [35]

$$\dot{z} = (\alpha + i\omega_0)z - |z|^2 z + \mathcal{F}(t), \qquad \gamma = -\alpha > 0, \tag{31}$$

describes a complex mode z with damping γ and natural angular frequency ω_0, driven by the applied input $\mathcal{F}$. For a weak input $\mathcal{F} = \varepsilon f(t)$ on the stable side, the perturbative response has the form $z = \varepsilon z_1 + \varepsilon^3 z_3 + \cdots$. There is no second-order source: the first-order response contains only the applied frequencies, and the cubic first contributes at order ε^3. Thus the normal form has a resonant linear susceptibility but lacks the quadratic difference-frequency generation of Eq. (10). This statement concerns the weak-input expansion; it is not an exclusion of every higher-order intermodulation or strong-drive response.

In the numerical control ($\alpha = -0.5$, $f_0 = 10$ Hz, $f_c = 100$ Hz, $\varepsilon = 0.5$), a square-law detector gives a beat-frequency lock-in amplitude of 2.5×10^{-1}, while $\mathrm{Re}\, z$ gives approximately 2×10^{-8}, at the numerical floor. Driving the same mode directly in band recovers the forced $1/\gamma$ growth. The control separates resonant amplification from the nonlinear source required to drive it at the beat.

D Efficiency of a memoryless transfer against a direct in-band drive

For the logistic family of memoryless transfers, compare balanced TI tones of polarization amplitude ε each with a direct in-band drive of amplitude 2ε, matching peak polarization. Their leading drive amplitudes are $|\sigma''|\varepsilon^2/2$ and $2\sigma'\varepsilon$, so $\eta_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$. This ratio concerns the specified transfer family and input matching, not TI in general or the dynamic detector of §2.3.

Writing $s = \sigma/(2e_0) \in (0,1)$ gives $\sigma' = 2e_0\rho s(1-s)$ and $\sigma'' = 2e_0\rho^2 s(1-s)(1-2s)$. Since $\mathrm{PEIX} = 4s(1-s)$, the leading ratio satisfies

$$\eta_{\mathrm{det}} \simeq \frac{\rho\,\varepsilon}{4}\sqrt{1-\mathrm{PEIX}} \;\le\; \frac{\rho\,\varepsilon}{4}. \tag{32}$$

It vanishes at the inflection, where the quadratic coefficient is zero, and approaches its bound in the saturated tails. This maximizes a ratio rather than an absolute response: both drives vanish in those tails. For the JR slope $\rho = 0.56\ \mathrm{mV}^{-1}$, the perturbative bound is 1.4% at $\varepsilon = 0.1$ mV and 0.28% at $\varepsilon = 0.02$ mV.

The slope is itself a parameter, so the question is what the whole sigmoid family can do. The logistic transfer depends on voltage only through $\rho(v - v_0)$, so writing

$$u = \rho\,\varepsilon, \qquad x = \rho\,(v^* - v_0), \tag{33}$$

we have $\sigma(v^* + \varepsilon q(t)) = 2e_0\,L(x + u\,q(t))$ for any normalized waveform q, with $L(z) = (1+e^{-z})^{-1}$. Every Fourier amplitude, and hence their ratio, is a function of (u, x) alone; the factor $2e_0$ cancels. Maximizing over v^* and ρ at fixed ε is therefore the same as maximizing over (u, x), and the family supremum is independent of field strength. Evaluating the exact cycle response rather than its quadratic truncation and optimizing x at each u, the ratio rises monotonically over the range explored and converges to the hard-threshold limit

$$\sup_{\rho,\,v^*} \eta_{\mathrm{det}} = 0.3479, \tag{34}$$

approached as $\rho \to \infty$ at a threshold offset δ (the distance from the operating point to the inflection) with $\delta/2\varepsilon = 0.523$. There is no finite optimal transfer width: narrower is always better, and reaching 95% of the ceiling requires $u \gtrsim 6$, that is $1/\rho \lesssim \varepsilon/6$.

Such a detector is not weak. At the optimum, and at the Vieira-scale polarization $\varepsilon \simeq 0.1$ mV, its open-loop beat drive exceeds the physiological-slope sigmoid's by some three orders of magnitude for the same e_0 and waveform, because the field excursion is about twice the threshold offset and the transfer switches across its full range instead of responding linearly. The disparity grows as the field weakens, the smooth sigmoid's response falling as ε^2 while a threshold repositioned with the field keeps its excursion.

The obstacle is the width. At Vieira et al.'s fields the ceiling is approached only for $1/\rho \lesssim 17\ \mu$V, and at human polarizations for $1/\rho \lesssim 3\ \mu$V, with the threshold offset held to the same order as the applied polarization. The width of a neural-mass sigmoid is set by the spread of thresholds and noise across $\sim 10^4$ neurons and is of order millivolts. A microvolt width lies far below that and reads as an idealized microscopic threshold or rectifier rather than as a neural-mass sigmoid; real thresholds are themselves noisy and state-dependent, so it is not a measured single-cell property either. The regime in which the family approaches its ceiling is thus one in which the "population sigmoid" has ceased to describe a population.

The limit is a step detector, not an envelope follower. For a unit step at offset δ the carrier-averaged transduction is $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and zero below; with $\beta = \delta/2\varepsilon$ its tACS fundamental is $(2/\pi)\sqrt{1-\beta^2}$ and its TI coefficient reduces by parts to $\eta_{\mathrm{det}} = 2\beta\,[K(m) - E(m)]/(\pi\sqrt{1-\beta^2})$ with $m = 1-\beta^2$ and K, E the complete elliptic integrals, giving 0.3479 at $\beta = 0.523$ by direct quadrature and by the elliptic form alike (`threshold_bound_check.py`). At that optimum the

step's second-to-first harmonic magnitude ratio is 0.151, not the 1/5 of a linear envelope follower. A bounded nondecreasing transfer is a positive mixture of shifted steps and both Fourier coefficients are linear in the transfer, so the step optimum bounds the family; the maximum itself is a numerical evaluation, not a certified bound.

E Extension: the laminar column at low carriers

This extension drives the two-band laminar column (LaNMM, §2.1) with carriers of 25–80 Hz and polarizations up to 3 mV, outside the perturbative regime of the main text and below the carriers TI uses. It is included for what it shows about a band-limited intermediary between field and slow generator; it does not carry the kHz argument.

The single JR column isolates the mechanism; the laminar model (LaNMM, §2.1) confirms it in a cortical column with two resonances of its own. Driving one population with the two-tone field of Eq. (26) and reading the alpha-band power of the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials (Methods, Fig. S2), we recover the same envelope resonance, now carrying the column's intrinsic rhythms.

As a consistency check, driving the deep P_1 (alpha) loop directly reproduces the alpha-band beat resonance of the JR column on both the stable-focus and limit-cycle sides, with the sweep and its code in the repository; what the laminar model adds is the carrier-dependent structure below.

One qualification governs the laminar sweeps. They run to $\varepsilon = 3$ mV on P_2, above the sigmoid voltage scale $1/\rho \approx 1.8$ mV, so they sit deliberately in a nonlinear, non-perturbative regime. They illustrate nonlinear frequency selection, not a verification of the weak-field square law; that verification is the low-amplitude JR work (Fig. 6b), which fits the square law at 0.05–0.2 mV, one to two orders of magnitude below $1/\rho$.

Sweeping the field amplitude ε and beat frequency Δf at a fixed carrier produces a region of enhanced alpha-band power: the response in P_1 is largest when Δf matches the deep alpha ($\sim$10 Hz) and grows with ε (Figs. S1a,b and S7a,b). The dominant-frequency contours identify where the output is close to the beat, without establishing phase locking of the autonomous rhythm. Away from the enhancement region the drive *reduces* alpha-band power: with the field on P_1, 95% of the amplitude–beat grid and 97% of the carrier–beat grid lie below the field-free baseline (Fig. S7), the drive pulling the autonomous alpha out of the 8–12 Hz band or suppressing it; with the field on P_2 the fractions are 44% and 31% (Fig. S1).

The carrier dependence illustrates the two coupling limits of §2.2–§2.6 within this model's 25–80 Hz carrier range. When the field is delivered to the superficial loop (P_2), alpha-band power enhancement in P_1 is largest in the gamma carrier band with Δf near the P_1 alpha (Fig. S1c,d). At $\Delta f = 10$ Hz, the carrier maximum is 44 Hz. This selectivity is consistent with a double resonance: the fast loop acts as a tuned radio-frequency (RF) front end whose dynamics affect the carrier dependence of the deep response. With direct P_1 drive, a ridge near the column's own alpha persists across the carrier range (Fig. S7c). That ridge contains autonomous activity and is not a measurement of the demodulated component alone.

The lock-in also contains the autonomous rhythm when the beat is close to alpha; its maximum over beats cannot isolate the driven response. We therefore compare the off-alpha statistic $S(f_c)$ defined in Methods (Table 6). With P_2 drive it peaks near 39 Hz and falls to about 16–17% of that peak at a 25 Hz carrier. With P_1 drive it peaks at 25 Hz and spans about 42–68% of that peak over 30–80 Hz. These are properties of the specified amplitudes and beat average, not universal carrier-rejection factors. Single-beat profiles have deeper minima, including a dip near a 45 Hz carrier under direct P_1 drive: the gamma circuit can affect the response in either stimulation condition.

The carrier peaks and aggregate ranges persist with the longer window, although individual normalized values change by up to 3.1 percentage points for P_2 drive and 9.6 for P_1 drive. The smallest S exceeds its mean field-free background by factors of about 16 and 8, respectively, with the 3 s window, and 62 and 32 with 10 s. Some individual beat–carrier points remain near or below background; the comparison concerns the mean over beats, not each grid point. TI's kHz carriers lie outside this model's range. Within it, the carrier can excite a network mode and affect

Table 6: LaNMM carrier profiles measured away from alpha. $S(f_c)$ is the mean lock-in amplitude of v_{P_1} over the 17 beats specified in Methods. Percentages are normalized to each row's own carrier maximum; the range and median use carriers from 30 to 80 Hz.

Drive	Window	Peak S (mV)	Peak f_c (Hz)	Range (%)	Median (%)
P_2, 1.5 mV	3 s	0.450	39	23–100	42
P_2, 1.5 mV	10 s	0.450	39	23–100	42
P_1, 0.5 mV	3 s	0.092	25	42–68	53
P_1, 0.5 mV	10 s	0.091	25	42–67	52

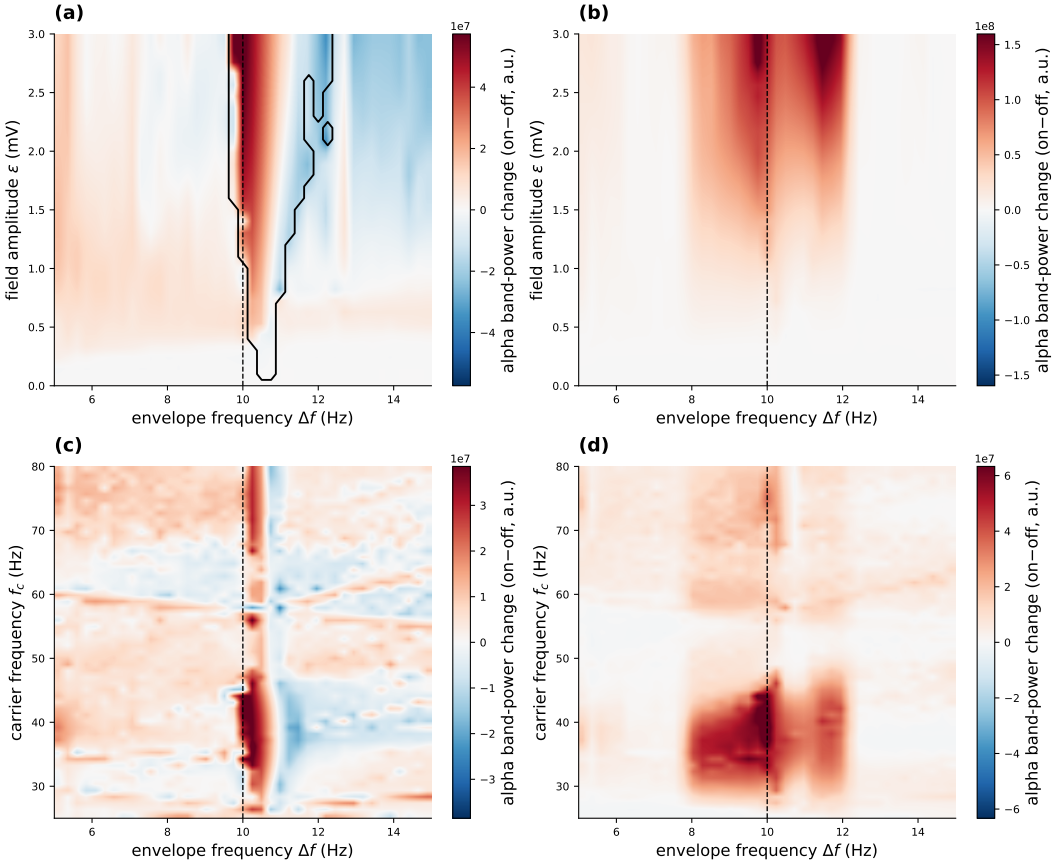


Figure S1: LaNMM frequency selection: two-tone TI field on the superficial P_2 (gamma) loop. The field enters the P_2 sigmoid argument as a membrane polarization ε (mV) with all external drives flat. Color shows the signed change of alpha-band (8–12 Hz) power from the unstimulated $\varepsilon = 0$ baseline, on a zero-centered scale (arbitrary units). *(a,b)* P_1 and P_2 enhancement vs. ε and Δf (carrier 40 Hz), peaking at the P_1 alpha (10.0 Hz, dashed). The black contour in *(a)* is the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), drawn because band power alone cannot separate a driven from an autonomous rhythm; it is a dominance screen, not a phase-capture test. The $\varepsilon = 0$ row is excluded, since with no drive the autonomous 10.67 Hz alpha already lies within the tolerance of two grid points (a resolution floor of 0.5 Hz); the set widens to 2.75 Hz at 3 mV. *(c,d)* Enhancement vs. carrier and Δf at $\varepsilon = 1.5$ mV: P_1 alpha-band enhancement is largest in the gamma carrier band with Δf near the P_1 alpha. At $\Delta f = 10$ Hz, the carrier maximum is 44 Hz. The field-free P_2 gamma peak is near 39.7 Hz; the finite-drive maximum need not coincide with it.

the detected response, qualifying the inert-carrier limit of §2.5 and linking it to frequency-tagging and intermodulation paradigms [57].

The LaNMM predicts that TI efficacy should (i) peak when Δf matches a region's intrinsic alpha; (ii) widen and strengthen with modulation amplitude; and (iii) for fields delivered via superficial

fast circuits, be largest when the carrier is matched to a gamma network resonance, for example two nearby gamma carriers summing to a ∼40 Hz-centered AM field, while remaining present, weaker, for off-resonance carriers. The third is untested: existing TI studies of deep targets sweep the *beat* frequency rather than the carrier against a network resonance [58, 59], so they neither support nor contradict it.

F Vocabulary and metrics

Forced response. Below the Hopf the column does not oscillate on its own; we quantify its response by the lock-in amplitude at the beat Δf (Eq. (25)). There is no autonomous rhythm to entrain in the stable-focus J-curve, timing, and tACS sweeps.

Frequency dominance and entrainment. Above the Hopf the column free-runs at f_0. A 1:1 phase-capture test would follow the unwrapped phase difference from the drive and require absence of secular drift. Our maps instead report $|f_{\text{out}} - \Delta f| < 0.3$ Hz, where f_{out} is the largest non-DC FFT peak. This is a frequency-dominance screen: a forced component can outgrow a persisting autonomous one without capturing it. When the components coexist, instantaneous-phase estimates also require their separation [60]. The maps therefore identify a dominance region, not a measured Arnold tongue.

Beat-locked rate and gating. Gating means that the amplitude of a persisting rhythm is modulated by the beat without frequency capture [36, 37]. The QIF statistic used here is the lock-in of the population rate r_E at Δf, complemented by the rate folded on beat phase over whole cycles. It measures beat-locked firing. A raw rate fold does not extract the gamma amplitude, and sidebands can arise from amplitude or phase modulation. The above-Hopf QIF panels therefore establish coexistence and mixing of the beat and gamma rhythms; a gamma-amplitude estimate would be needed to identify gating specifically.

The protocol for Fig. S4 places the column on the limit-cycle side (input p between the saddle-node on invariant circle and the Hopf), drives it with $s(t) = \varepsilon \cos \Omega t$, and records the dominant frequency. Grid widths and the clipping check are specified in the code. As the Hopf is approached, the autonomous cycle shrinks and the frequency-dominance region broadens in these simulations. Its width and the stable-focus susceptibility describe different responses; their increase near onset does not make them the same quantity.

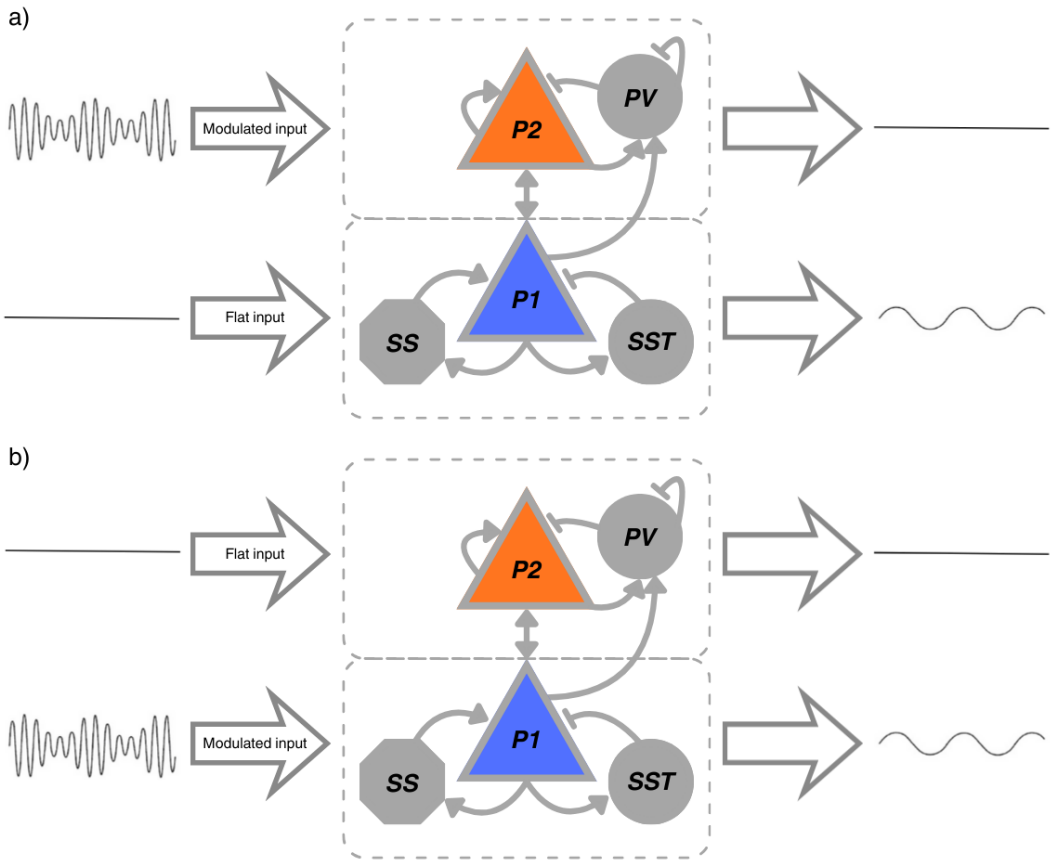


Figure S2: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the TI field drives P_2 while P_1 receives a flat drive; *(b)* the TI field drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the generated beat-frequency component is transferred to the deep loop: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency component (right traces).

G Supplementary figures

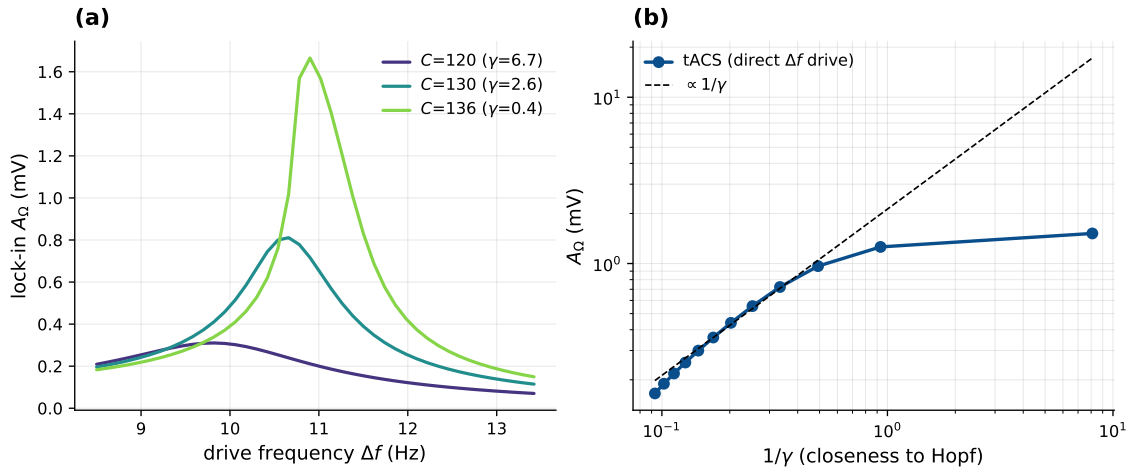


Figure S3: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the field-free operating point pinned, sweeping the coupling $C\,(=J)$ toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 9). The leading tACS response is linear in the field; the TI beat is quadratic. Their response ratio is independent of damping in linear response and changes at the finite amplitudes nearest the Hopf (§4.6). Here $\gamma > 0$ describes the field-free state.

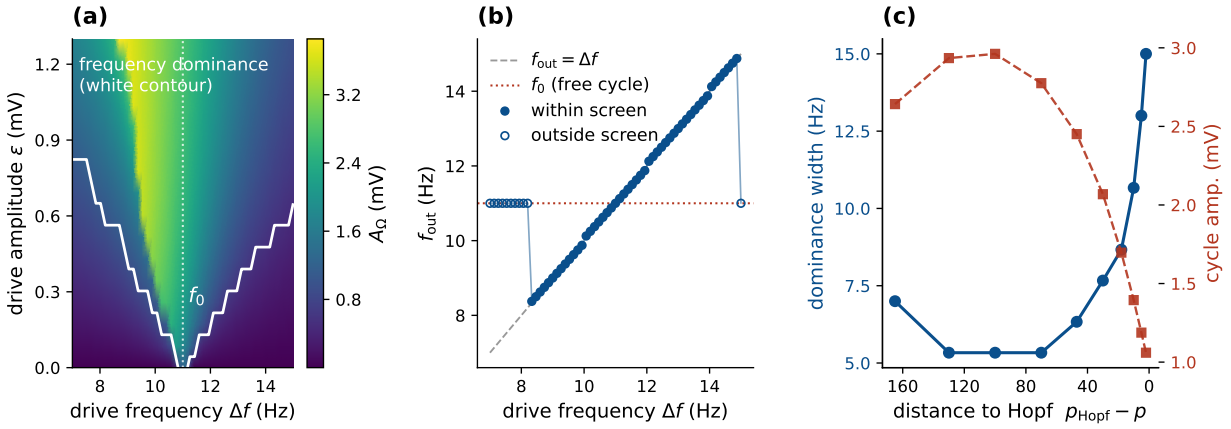


Figure S4: Frequency dominance in the driven autonomous Jansen–Rit column. The column is set in its autonomous alpha limit cycle ($p = 250$, $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid. *(a)* The set $|f_{out} - \Delta f| < 0.3$ Hz forms a wedge near f_0 that widens with drive. It screens for phase capture but does not establish it. *(b)* Inside this region the dominant frequency follows Δf; outside, it remains near f_0. *(c)* Toward the Hopf at fixed drive the dominance width grows as the autonomous cycle amplitude vanishes (blue, left axis; red dashed, right axis). Above the Hopf this width describes frequency selection in a driven autonomous oscillator; it is distinct from the linear susceptibility of the stable-focus side (Fig. 9). Both diagnostics increase toward criticality in these sweeps.

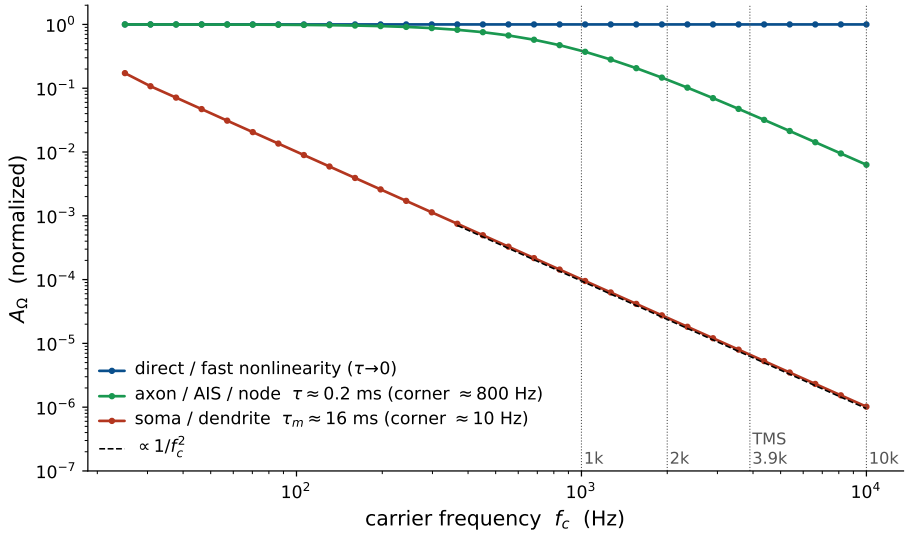


Figure S5: Carrier attenuation depends on the assumed access time (Jansen–Rit, open loop). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide); the assumed fast response ($\tau = 0.2$ ms) retains about 0.14 of the direct response at 2 kHz and 0.04 at 3.9 kHz. These are sensitivity cases, not measurements of anatomical compartments.

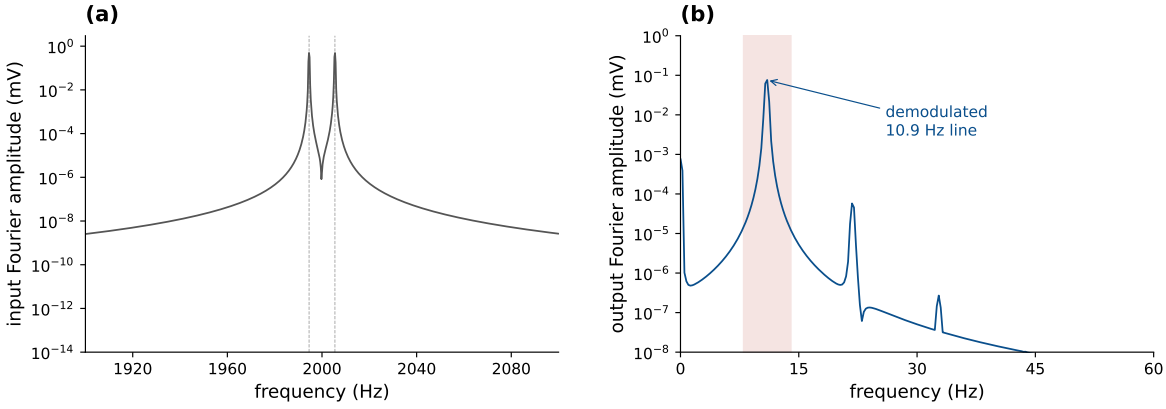


Figure S6: Two-tone TI at a kHz carrier (Jansen–Rit, assumed fast access). Two equal tones at $f_1 = 1994.54$ and $f_2 = 2005.46$ Hz, beating at approximately 10.9 Hz, pass through the assumed low-pass ($\tau = 0.2$ ms) into the recurrent column. *(a)* The input amplitude spectrum has lines only at f_1 and f_2; in the 8–14 Hz band its maximum is 2.2×10^{-13} mV. *(b)* The output carries a demodulated line at the beat, 10.9 Hz.

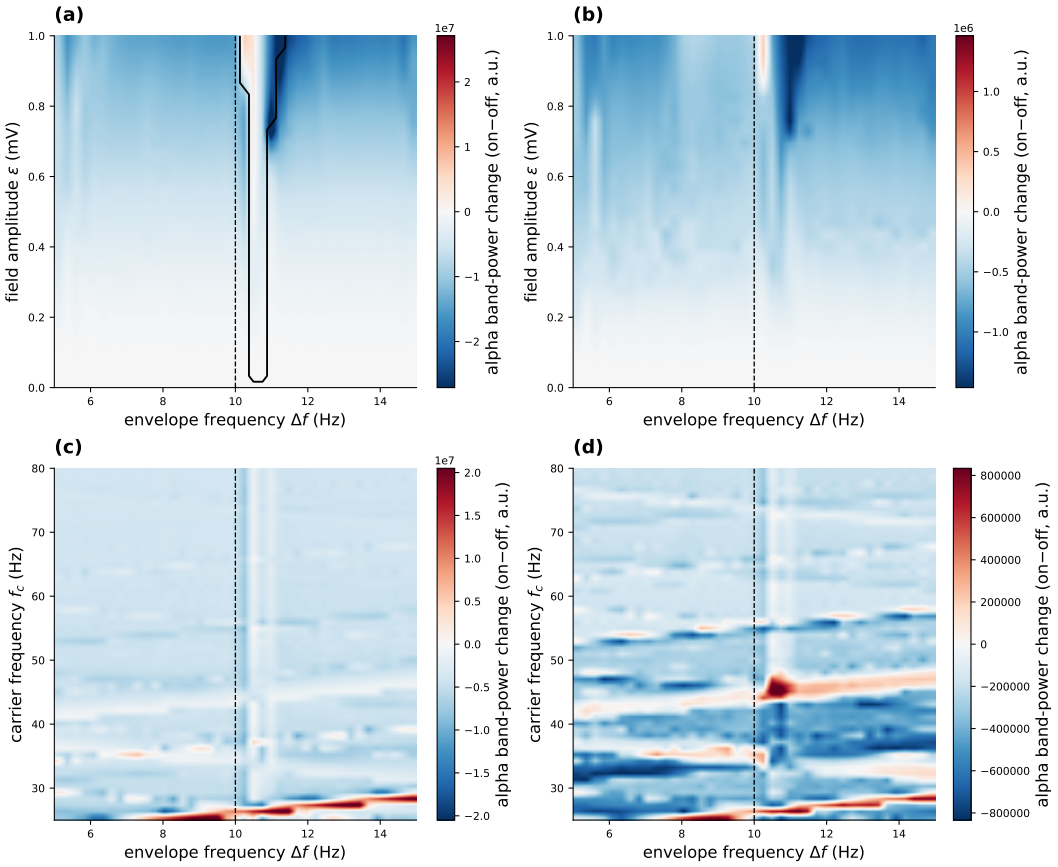


Figure S7: LaNMM frequency selection: two-tone TI field delivered directly to the deep P_1 (alpha) loop. As Fig. S1 but with the field on P_1. The deep loop shows frequency dominance at smaller fields, so the amplitude range is $\varepsilon \leq 1$ mV here against 3 mV there. These finite-amplitude maps extend beyond the perturbative regime of the theory. Color: signed change of alpha-band power from the $\varepsilon = 0$ baseline, zero-centered; most of the grid is suppression. *(a,b)* P_1 and P_2 enhancement vs. ε and Δf; the black contour in *(a)* outlines the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), computed from the dominant output frequency because band power alone cannot separate a driven from an autonomous rhythm; the $\varepsilon = 0$ row is excluded (resolution floor 0.5 Hz). The set widens to 1.25 Hz at $\varepsilon = 1$ mV. *(c,d)* Enhancement vs. carrier frequency and Δf at $\varepsilon = 0.5$ mV. The ridge near alpha persists across the carrier band and includes the autonomous rhythm. A separate lock-in comparison away from alpha, including its window and background controls, is given in Table 6; these panels show band-power changes.

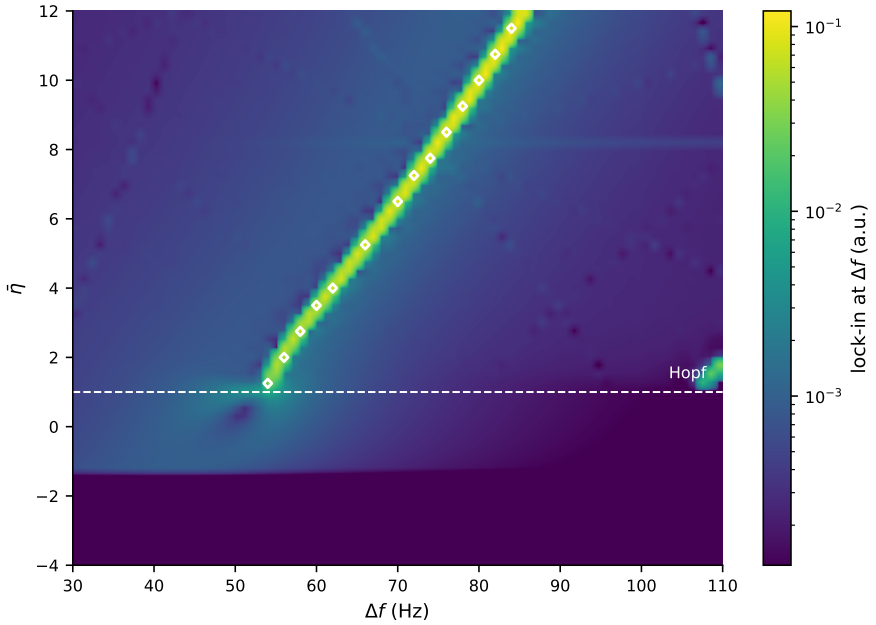


Figure S8: NMM2 PING resonance map (exact mean field). Demodulated response over the (Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency, with f_0 rising with $\bar{\eta}$ for these parameters. White markers outline the frequency-dominance set (Appendix F), drawn only above the Hopf (dashed line) where there is an autonomous rhythm to capture; below it the same color field is a forced response. The markers do not establish phase capture. Lock-in amplitude cannot separate driven and autonomous contributions, since both can place power at Δf.

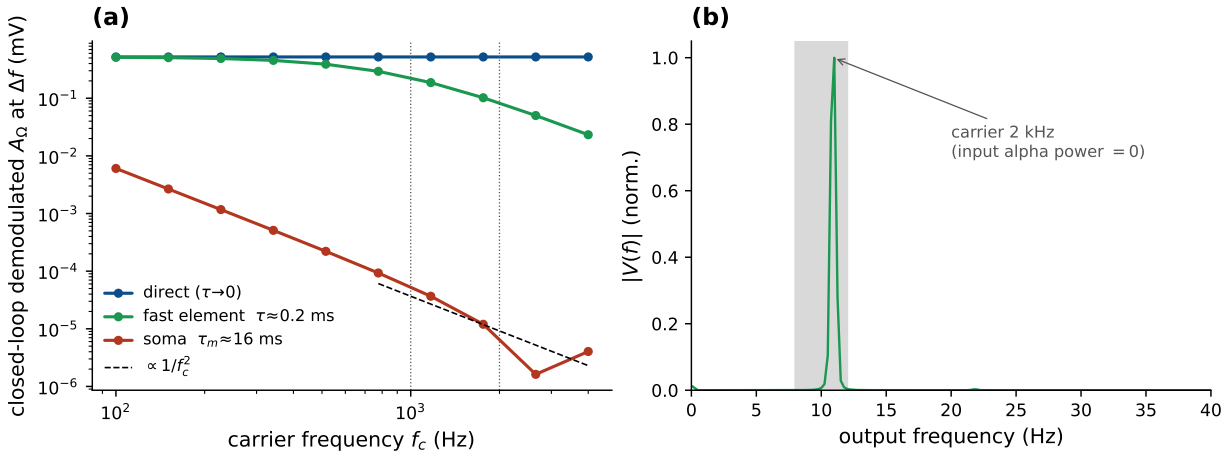


Figure S9: Demodulation at a kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three access models: direct ($\tau \to 0$, carrier-independent), fast ($\tau = 0.2$ ms), and somatic ($\tau_m = 16$ ms; $1/f_c^2$ roll-off). The fast time is assumed rather than assigned to a measured compartment. *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: an alpha line at f_0, synthesized by the column although the input carries no alpha power.

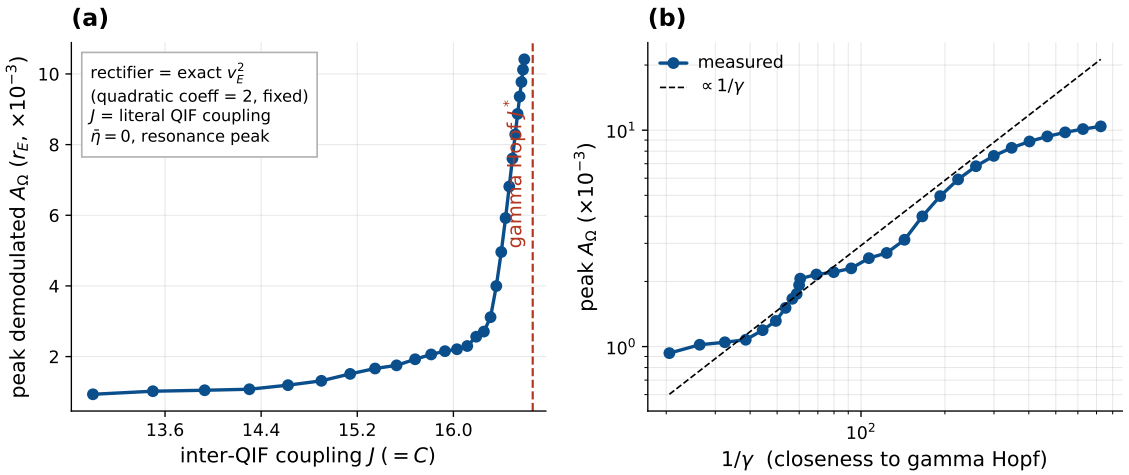


Figure S10: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). $J\,(= C)$ is the literal inter-QIF synaptic weight. The sweep is unpinned and uses the leading fast-carrier source for normalization (§2.3); its finite-carrier corrections have not been bounded across the full sweep. The plotted Hopf and damping γ refer to the field-free equilibrium. The full forced endpoint is stable (largest multiplier 0.034 over 2 s), despite the instability predicted by a DC-only averaged equilibrium. *(a)* The peak demodulated response of r_E grows as $J \to J^*$ (gamma Hopf, $\bar{\eta} = 0$; field-free stable focus, $\gamma > 0$). *(b)* The response is compared with the $1/\gamma$ reference (dashed), with finite-amplitude departures near criticality. Both coupling and rectifier are derived from the QIF network.

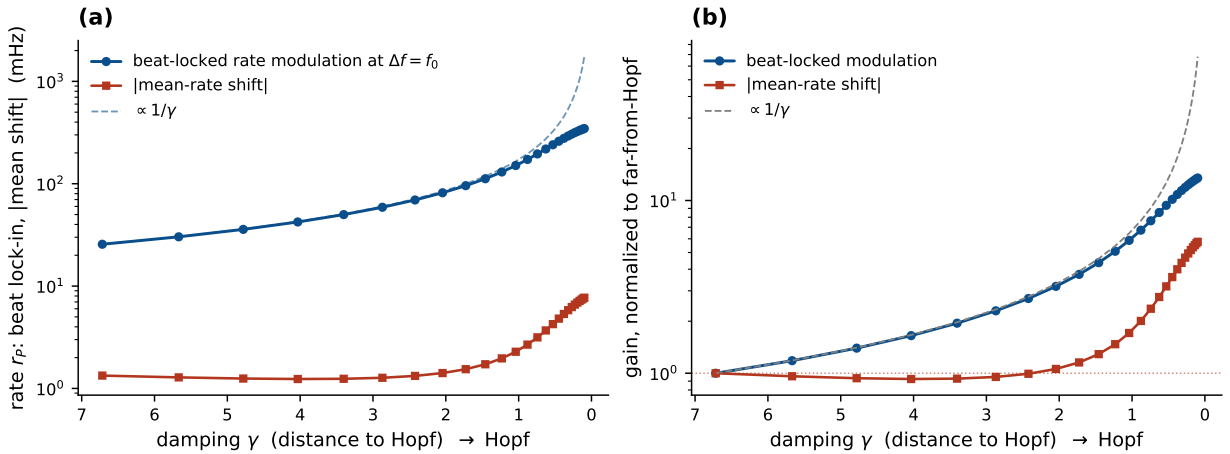


Figure S11: Envelope-locked modulation against mean-rate shift (Jansen–Rit, detection pinned). As the column is driven toward the Hopf (from the stable side, $\gamma > 0$) through the coupling at fixed $\sigma''(v^*)$, both quantities are read from the pyramidal rate $r_P = \sigma(v + s)$, in mHz. *(a)* The beat-locked modulation at $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed) before saturating; the magnitude of the mean-rate shift (red) stays at a few mHz and rises near the Hopf. *(b)* Both normalized to their far-from-Hopf values (log scale): the modulation gain follows $1/\gamma$; the mean-shift gain rises later and less. Modulation exceeds mean shift by 19× far from the Hopf and 45–66× near it.

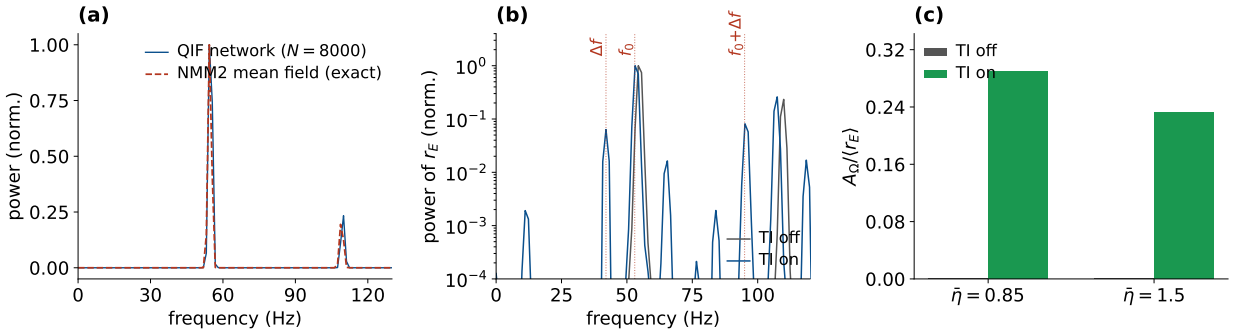


Figure S12: Beat-locked modulation and mean-rate change in the QIF network. *(a)* Field-free power spectrum of $r_E(t)$, normalized: the spiking network ($N = 8000$) matches the exact NMM2 mean field in gamma frequency (∼55 Hz); a normalized spectrum does not test the mean rate. *(b)* Intermodulation above the Hopf (TI off vs. on, log power): the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the persisting gamma near 53–54 Hz, raising the direct Δf line and the sum sideband near 95 Hz, both absent with TI off. These sidebands establish nonlinear mixing; they do not uniquely identify gamma-amplitude gating. *(c)* The Δf modulation depth $A_\Omega/\langle r_E \rangle$, off vs. on: below 0.003 rising to 0.29 ($\bar{\eta} = 0.85$) and 0.23 ($\bar{\eta} = 1.5$), while the mean rate rises by about 8–11%. The labels identify field-free states; both field-on responses contain gamma activity beyond the beat component. Spectra use the same post-settling windows as the rate measurements and are normalized to each spectrum's maximum. Modulation depth is reported rather than the off→on lock-in ratio, whose denominator is the field-free floor.

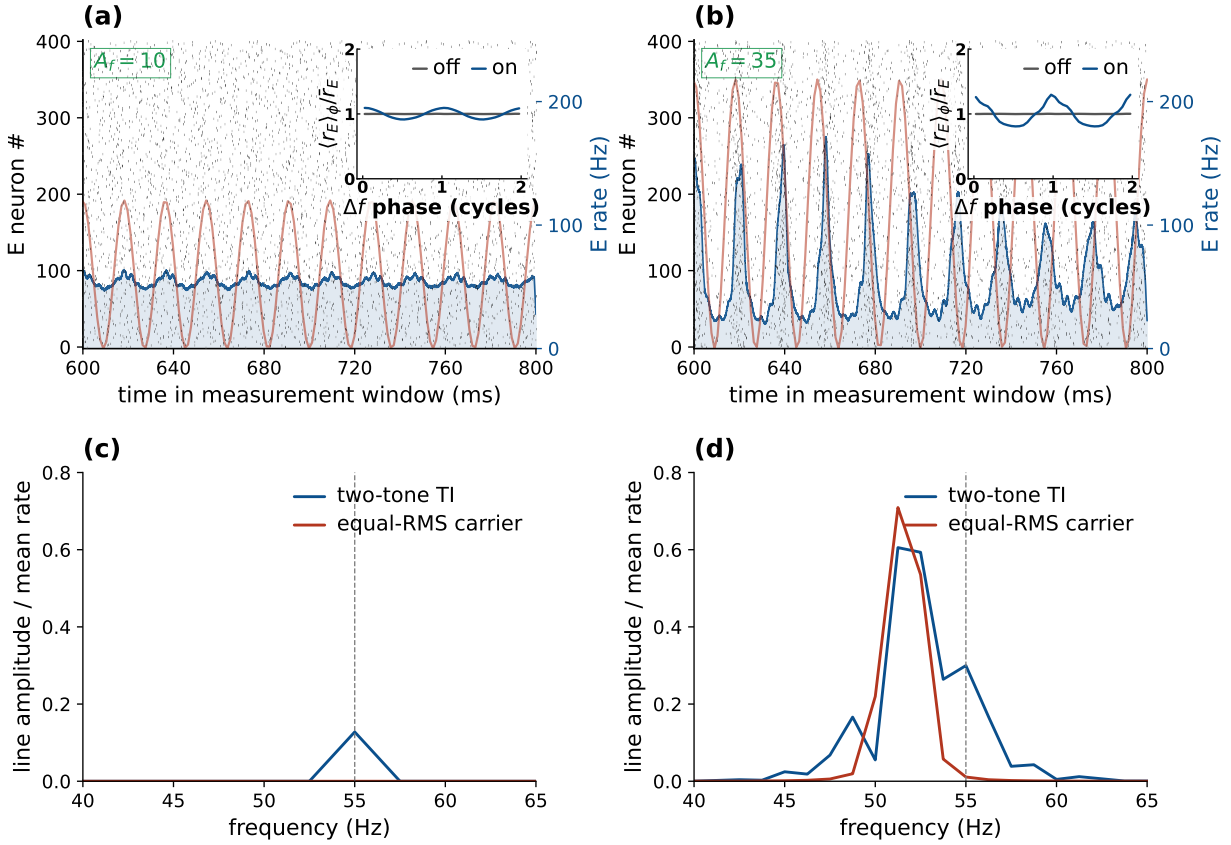


Figure S13: Beat modulation below and across the induced-gamma transition. All panels use $\bar{\eta} = 0.85$, carrier 300 Hz, and TI beat 55 Hz. *(a,b)* Spiking-network responses at $A_f = 10$ and 35 per tone, respectively: black ticks are spikes from 400 of 8000 E neurons; blue is the smoothed population rate on the common right-axis scale, and red is the envelope in arbitrary units. Insets show phase-folded rate divided by its mean (off gray, on blue), on the same scale. The beat depths are 0.093 and 0.290, and the mean-rate increases are 2.8% and 10.5%. At the larger amplitude, non-beat-locked gamma dominates the raster and is suppressed by the fold. *(c,d)* Corresponding mean-field spectra under TI (blue) and an equal-RMS single carrier (red), whose peak amplitude is $\sqrt{2}A_f$. Line amplitude is normalized by each trace's mean; the dashed line marks 55 Hz. Spectra use a Hann window over 800 ms after 800 ms settling. Both waveforms induce gamma at the larger amplitude, while TI produces the additional beat response. The carrier's apparent 55 Hz line falls below 10^{-4} with a 3.2 s window, while TI's persists. The carrier comparison is a mean-field result. The spiking panels retain $V_{\text{peak}} = 100$; at cutoffs 200–400, the $A_f = 10$ depth ranges from 0.121 to 0.141 across the two tested steps (§4.8).